TRADI-WIN: A Trustworthy, Model-Choosing, Model Context Protocol-Driven Agentic System for Explainable Crypto Trading

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List of Abbreviations

Abbreviation	Description
ADA	ADA (native cryptocurrency of the Cardano network)
AI	Artificial Intelligence
ALGO	ALGO (native cryptocurrency of the Algorand network)
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
ATOM	ATOM (native cryptocurrency of the Cosmos Hub network)
AVAX	AVAX (native cryptocurrency of the Avalanche network)
BNB	BNB (native cryptocurrency of the BNB Chain network)
BTC	Bitcoin (native cryptocurrency of the Bitcoin network)
CPU	Central Processing Unit
CSV	Comma-Separated Values
DM	Diebold–Mariano (forecast comparison test)
DOT	DOT (native cryptocurrency of the Polkadot network)
ETH	Ether (native cryptocurrency of the Ethereum network)
FinBERT	Financial Bidirectional Encoder Representations from Transformers (finance-domain BERT model)
FTM	FTM (native cryptocurrency of the Fantom network)
GRU	Gated Recurrent Unit
JSON	JavaScript Object Notation
JSONL	JSON Lines (newline-delimited JSON format)
LINK	LINK (native cryptocurrency of the Chainlink network)

LLM	Large Language Model
LSTM	Long Short-Term Memory (recurrent neural network)
LTC	Litecoin (native cryptocurrency of the Litecoin network)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MATIC	MATIC (native cryptocurrency of the Polygon network)
MCP	Model Context Protocol
MCP-Tradi-Win	Proposed MCP-driven modular multi-agent crypto-trading framework
MIXRTs	Mixing Recurrent Soft Decision Trees
RMSE	Root Mean Squared Error
RW	Random Walk (baseline forecaster)
SES	Simple Exponential Smoothing (baseline forecaster)
SOL	SOL (native cryptocurrency of the Solana network)
SOPR	Spent Output Profit Ratio
TFT	Temporal Fusion Transformer
TVL	Total Value Locked
UCT	Coordinated Universal Time
UI	User Interface
UNI	UNI (governance token of the Uniswap protocol)
USD	United States Dollar
VDN	Value Decomposition Networks

XAI	Explainable AI
XRP	XRP (native cryptocurrency of the XRP Ledger)

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Abstract

Reliable and transparent cryptocurrency forecasting remains challenging due to market volatility and the opacity of many machine learning models. Results demonstrate that a transparent and verifiable multi-agent trading framework, coordinated through a Model Context Protocol (MCP), can improve forecasting reliability and decision accountability**.** The system combines statistical and neural network predictions with sentiment-adjusted trade thresholds to generate explainable trading signals. A custom model context protocol (MCP) orchestrates specialized agents responsible for data collection, prediction modeling, emotional tone evaluation, and trade execution through unified context sharing, standardized communication, and traceable operations. The data collection agent gathers cryptocurrency valuations and media coverage from open Application Programming Interfaces (APIs). The prediction agent compares AutoRegressive Integrated Moving Average (ARIMA), Prophet, a Long Short-Term Memory (LSTM) neural network trained on return sequences, and two reference methods; a naive Random Walk (RW) and a Simple Exponential Smoothing (SES).

The best model is then selected via a weighted performance metric. This metric combines 40 percent directional prediction accuracy, 30 percent of one minus Mean Absolute Percentage Error (MAPE), and 30 percent of one minus the Mean Absolute Error (MAE)-to-price ratio. Emotional tone is derived from current cryptocurrency news headlines through NewsAPI and TextBlob, enhanced with trading volume and temporal weighting factors, with optional FinBERT validation. The trade execution logic modifies the confidence threshold by ten percent upward or downward depending on whether sentiment supports or contradicts the prediction. Trade signal confidence is calculated as 60 percent directional accuracy combined with 40 percent of one minus the Root Mean Squared Error (RMSE)-to-price ratio for the chosen predictor. Large Language

Models (LLMs) serve solely to generate human-readable explanations and are restricted to referencing only the analytical outputs created by the framework itself, thereby improving reliability and minimizing unsubstantiated assertions.

Using Ether daily data from August 17, 2024 to August 16, 2025 (365 days), the system achieved a weighted performance score of 87 percent for ARIMA and a signal confidence of 81.1 percent, surpassing the sentiment-adjusted threshold of 70 percent to produce a definitive trading recommendation.

Contributions include an auditable MCP-coordinated multi-agent framework, a composite model-selection metric tied to directional accuracy and scale-normalized errors, and a sentiment-adjusted confidence gate with explanation-only LLM usage; implications are a reproducible blueprint for trustworthy, metric-driven deployment and an audit trail suitable for risk and compliance review.

Keywords: Multi-Agent Systems, Model Context Protocol, Cryptocurrency Trading, Large Language Models, Trustworthiness

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Chapter 1: Introduction

Cryptocurrency marketplaces are highly volatile, globally distributed, and continuously influenced by news, sentiment, and algorithmic trading. Automated forecasting systems deployed in these environments must not only perform well but also provide clear, auditable decision traces to maintain trust among traders, institutions, and regulators. However, many existing AI-driven financial systems rely on opaque model pipelines that limit interpretability and complicate oversight, creating a need for architectures that combine predictive accuracy with transparent decision reasoning.

Artificial intelligence (AI) refers to computational methods that learn from data to perceive, reason, and decide. Multi-agent systems approach complex problems by assigning different roles to separate agents that observe, plan, and act within a shared environment, which improves scale, fault tolerance, and adaptability under changing conditions (Li et al., 2024). Large language models (LLMs) trained on very large text collections can interpret and generate language and code, and within agent-based workflows they can plan tasks, call tools, and synthesize knowledge across components (Shanahan, 2024). Model Context Protocols (MCP) provide shared rules for exchanging context, memory, and reasoning records so one component can reliably use and audit the outputs of another. Combined, these ideas create transparent and reproducible decision pipelines and form the operational basis of MCP-Tradi-Win.

MCP-Tradi-Win refers to the proposed system. MCP denotes the orchestration layer that standardizes context exchange, tool calls, and reasoning traces across agents. “Tradi” indicates the framework’s focus on trading tasks, while “Win” underscores its objective: enabling well-

grounded decisions with end-to-end transparency suitable for stakeholder review and regulatory audit.

In finance, these capabilities can improve accuracy and trust while reducing operational exposure. Forecasting agents can deliver signals for short, medium, and long horizons, sentiment agents can convert news and social data into structured indicators, and an LLM planner can coordinate tasks and resolve conflicting model outputs. MCP allows consistent information flow across pricing, risk, and execution services and preserves audit trails for compliance, model risk oversight, and after trade analysis. The result can be faster scenario evaluation, better stress testing and regime recognition, stronger fraud and anti-money laundering monitoring with clearer evidence, tighter spreads, improved fill quality, fewer false alerts, quicker incident handling, and lower control costs with explanations that supervisors and stakeholders can review (Jensen et al., 2023

1.1 Background and Motivation

The digital currency market is characterized by high volatility, fragmented information sources, and frequent structural changes, all of which complicate dependable automation (Pessa et al., 2023). Most existing systems rely on single model frameworks that process limited forms of data, typically historical pricing (Fieberg et al., 2025). This limitation has become increasingly problematic as algorithmic trading expands into decentralized and high-frequency financial environments where accountability and real-time adaptability are essential.

At the same time, there is growing demand for financial AI systems that are transparent, auditable, and trustworthy (Černevičienė & Kabašinskas, 2024). In complex decision pipelines, opaque machine learning architectures erode user confidence, hinder oversight, and present

compliance risks. Prior research has emphasized that explainability must be embedded into the system's design rather than applied retrospectively. A transparent architecture allows forecasts, deviations, and sentiment indicators to be traced, verified, and reproduced, creating a verifiable chain of reasoning behind every decision (Yeo et al., 2025). Within this context, the proposed work is motivated by the need for an architecture that integrates both predictive accuracy and interpretability from the ground up.

This thesis introduces a modular multi-agent architecture coordinated by an MCP that unifies numerical and contextual data through standardized communication. Specialized agents handle forecasting, sentiment assessment, and decision synthesis, while an LLM functions as a constrained narrator that converts analytical records into human-readable explanations. By incorporating verification metrics such as confidence levels, deviation measures, and statistical validation tests, the system not only enhances forecasting precision but also provides a clear and reproducible decision trail. The aim is to demonstrate that financial automation can be both high-performing and accountable while meeting the dual needs of operational efficiency and regulatory transparency in the evolving landscape of cryptocurrency markets.

1.2 Problem Definition

The cryptocurrency market, referring specifically to decentralized digital assets such as Bitcoin, Ether, and other related tokens, is characterized by high volatility and unreliable, disorganized information channels, complicating dependable automation efforts. Numerous current solutions depend on single-model architectures that process limited data types, primarily historical pricing. These approaches face challenges incorporating subjective data and typically lack modularity or clarity, constraining comprehensibility and confidence (Weber et al., 2024). This research tackles these constraints by presenting a component-based multi-agent architecture

that merges numerical and contextual indicators while ensuring its choices are comprehensible and verifiable through intentional design.

1.3 Research Gap

Research Gap 1: Lack of Integrated, Verifiable Architectures

Most existing automated trading systems operate as closed pipelines that emphasize predictive performance but neglect architectural transparency. These single-model designs often obscure internal processes and make it difficult to audit decision traces, which limits accountability and trust in financial automation (Cheong, 2024). There remains a gap in frameworks that explicitly integrate verification, traceability, and reproducibility into their design, allowing researchers and regulators to evaluate not only what decision was made, but why it was made.

Research Gap 2: Insufficient Fusion of Quantitative and Contextual Intelligence

Traditional forecasting models rely heavily on numerical data such as price and volume, disregarding qualitative market drivers like sentiment, news, and event signals. Recent studies demonstrate that advanced deep-learning architectures can integrate multiscale and relational dependencies to capture complex cryptocurrency dynamics, highlighting the importance of combining numerical and contextual information (Zhou et al., 2025). A gap therefore exists in developing architectures that harmonize quantitative indicators with contextual sentiment and language-based data to improve situational awareness and adaptive response.

Research Gap 3: Limited Statistical Comparability and Model Governance

Current systems rarely include standardized methods for statistically comparing competing models or documenting those results in a reproducible format. The absence of transparent reporting

standards and unified evaluation protocols weakens reproducibility and limits oversight across financial AI systems (Cheong, 2024). Addressing this gap requires a standardized, protocol-driven framework that records, compares, and validates each model's output under unified experimental conditions.

1.4 Research Questions

Contemporary progress in Multi-Agent Reinforcement Learning (MARL) for financial transactions exhibits remarkable results, especially in rapid-frequency trading scenarios. Structures like Value Decomposition Networks (VDN) paired with Multi-Agent Proximal Policy Optimization (MAPPO) have proven effective at refining approaches in fluid markets (Lopez-Lira, 2025). Nevertheless, these architectures fail to offer visibility into individual agent influences on results or rationales behind particular transaction choices. This comprehensibility deficit restricts their wider deployment in fields like digital currency transactions, where confidence and accountability are fundamental requirements.

RQ1. *How can a multi-agent protocol architecture be designed to maintain verifiable decision trails and comprehensibility without sacrificing predictive performance in cryptocurrency trading environments?*

Multi-agent reinforcement learning constructs have been utilized to model cryptocurrency marketplace conduct, with agents tuned using actual pricing information such as Binance settlement values (Lussange et al., 2024). These constructs improve comprehension of system-wide market mechanisms but lack features that clarify or validate choice-generation workflows. Without integrated comprehensibility, such platforms cannot sufficiently enable transparent

choice-generation in unstable crypto settings, creating a significant void for component-based architectures that unite both forecast precision and rationale.

RQ2. *In what ways can numerical indicators and contextual sentiment data be effectively fused within an agent-based system to improve situational awareness and adaptability of crypto-market forecasts?*

Comprehensibility-centered MARL approaches have surfaced in alternative fields, including robotics and collaborative scenarios. Techniques like Mixing Recurrent Soft Decision Trees (MIXRTs) explicitly display logic pathways and facilitate enhanced understanding of multi-agent choices (Liu et al., 2025). However, these approaches have not been transferred to financial transaction uses. This separation underscores an ongoing research void: the nonexistence of field-tailored, comprehensible MARL approaches that reconcile forecast effectiveness with clarity in multi-agent cryptocurrency transaction platforms.

RQ3. *How can standardized, statistically grounded evaluation protocols such as composite scoring and DM testing enhance model governance, reproducibility, and stakeholder confidence in financial AI systems?*

Although previous investigation exhibits developments in multi-agent reinforcement learning for financial transactions, cryptocurrency implementations remain constrained by the absence of integrated comprehensibility and transparent logic. To bridge these voids, this thesis presents a novel MCP that orchestrates dedicated AI agents for prediction, emotional tone evaluation, choice-generation, and comprehensibility. By incorporating comprehensibility into the architecture directly through confidence metric calculations, deviation examination, and choice verification records, this platform not only enhances forecast effectiveness but also guarantees

clarity. LLM constructs are utilized solely as restricted narrators that transform internal analytical records into brief, human-comprehensible narratives, enriching clarity and practical utility while preventing fabrications. This method bridges the void by providing a component-based, verifiable architecture appropriate for crypto marketplace instability and oversight requirements.

1.5 Objectives and Scope

This thesis presents a multi-agent cryptocurrency transaction architecture orchestrated by a tailored model context protocol that sustains unified context, regularized communications, and replicable records. The structure incorporates four dedicated agents: a data collector that obtains and standardizes pricing information from Yahoo Finance and news items from NewsAPI; a Forecast agent that assesses AutoRegressive Integrated Moving Average (ARIMA), Prophet, a returns-oriented Long Short-Term Memory (LSTM) neural network, and reference comparators including Random Walk (RW), also called Naïve, and Simple Exponential Smoothing (SES). A model is chosen through a composite metric that weighs prediction direction correctness at forty percent, one minus Mean Absolute Percentage Error (MAPE) at thirty percent, and one minus Mean Absolute Error (MAE) over price at thirty percent. A sentiment agent that extracts polarity from news items using TextBlob, supplements it with volume and temporal indicators, monitors historical success proportion, and can be validated with Financial Bidirectional Encoder Representations from Transformers (FinBERT).

Lastly, a decision agent that combines indicators and implements a sentiment-modified transaction threshold, raising or lowering the baseline threshold by ten percentage points based on correspondence. Rolling mini retrospective tests and Diebold-Mariano (DM) evaluations enable comparative assessment. DM offers a rigorous evaluation of whether two prediction approaches possess equivalent forecast precision by examining the temporal sequence of their deviation

differences (for instance, squared deviations) across the identical assessment period. Simply stated, it reveals whether one model's demonstrated benefit is improbable to stem from chance fluctuation, typically condensed by the DM measure and its probability value. It fits financial prediction because it permits any choice-aligned deviation and compensates for autocorrelation when periods intersect. The coordinated workflows, utilizing the MCP context, produce JavaScript Object Notation (JSON), Comma Separated Values (CSV), and JavaScript Object Notation Lines (JSONL) records and preserve sequential records for clarity and accountability. Narratives for end operators are constructed by an LLM restricted to the architecture's internal records to reduce unverified assertions. The platform was prototyped in Google Colab and implemented using Streamlit for participatory examination tied to a designated date.

1.6 Methodology Overview

The platform is realized as a multi-agent architecture administered by a tailored MCP that sustains unified context, regularized communications and sequential records to guarantee coherence, comprehensibility and accountability throughout the workflow. An orchestrated sequence implements four agents: Data Fetcher, Forecast, Sentiment, and Decision.

The Data Fetcher agent obtains cryptocurrency pricing information from Yahoo Finance, standardizes temporal markers, imposes uniform structures for subsequent handling, and ties each execution to an explicit examination date to enable replication. Financial news items are incorporated within the Sentiment agent when obtainable. It standardizes temporal markers, imposes uniform structures for subsequent handling, and ties each execution to an explicit examination date to enable replication.

The Forecast agent combines statistical and deep learning methodologies and assesses ARIMA, Prophet, and two reference models (RW and SES); a returns-oriented LSTM is conditionally incorporated when activated. Model choice relies on a composite scoring mechanism that allocates weighted significance to transaction-pertinent measurements as specified: prediction direction correctness at 40 percent, one minus MAPE at 30 percent, and one minus MAE divided by price at 30 percent. Prediction direction correctness is further calibrated to disregard price fluctuations beneath zero-point one percent to diminish the effect of marketplace interference. Forecast certainty is calculated from Directional Accuracy alongside a price-standardized deviation element, prioritizing accurate directional predictions while penalizing magnitude-dependent deviations. Rolling mini retrospective tests produce recent extrapolated forecasts, and DM evaluations are utilized to contrast competing models and measure whether precision differences are statistically meaningful.

The Sentiment agent employs a Proxy/NewsHybrid methodology: it produces a price-oriented proxy and, when news items are obtainable through NewsAPI, supplements the indicator with volume and temporal markers; it monitors historical success proportion and can be validated with FinBERT.

The Decision agent, functioning under protocol direction, combines indicators from the prediction and sentiment elements and generates buy, sell, or hold suggestions. A transaction threshold modification of plus or minus ten percentage points is implemented contingent on whether forecast and sentiment indicators correspond or clash, thereby raising or lowering the baseline threshold in a clear and regulation-driven fashion.

Narratives are constructed by an LLM that is employed exclusively for human-comprehensible narration. To diminish unverified assertions, the model is restricted to citing only

internal analytical records, encompassing composite metrics, individual model measurements, sentiment diagnostics, retrospective test products, and significance evaluations. Each execution produces confidence metrics, choice logs, and a comprehensive record of intermediate calculations, alongside durable records in JSON, CSV, and in JSONL structures. This blueprint enables replicable assessment, examination, and oversight of both model selections and ultimate transaction suggestions.

1.7 Thesis Structure

This thesis is organized into seven more chapters, each addressing a specific component of the research process and system development.

Chapter 2: Literature Review

This chapter reviews previous studies related to artificial intelligence, multi-agent systems, large language models, and model context protocols with a focus on their application to financial forecasting and trading automation. It also discusses work on explainable and interpretable AI, identifying conceptual and technical limitations that the proposed architecture aims to overcome.

Chapter 3: System Requirements and Architecture Design

This chapter defines the functional and nonfunctional requirements of the proposed platform and presents the complete system architecture. It describes the purpose and interaction of the data fetcher, forecast, sentiment, and decision agents while explaining how the MCP ensures consistent context, structured communication, and reproducible analytical records.

Chapter 4: Implementation

This chapter explains the practical realization of the MCP-Tradi-Win prototype. It details the tools, programming environment, and libraries used for development, including Python, Google Colab, and Streamlit. The implementation of agent interactions, constrained large language model narration, and the generation of structured outputs such as JSON and CSV logs for transparency and validation are also discussed.

Chapter 5: Experimental Design and Evaluation Methodology

This chapter describes the datasets, performance metrics, and validation strategies used to assess the system. It explains the application of comparative tests such as the DM evaluation, the computation of directional accuracy and deviation-based metrics, and the methods used to measure interpretability through traceability and narrative analysis.

Chapter 6: Results and Analysis

This chapter presents and interprets the results obtained from experimental runs. It compares the MCP-Tradi-Win framework with baseline models in terms of predictive accuracy, reliability, and explainability, and analyzes the interactions among agents to identify performance patterns and insights relevant to financial forecasting.

Chapter 7: Discussion

This chapter evaluates the broader implications of the findings in relation to the research questions. It analyzes how the MCP and multi-agent coordination contribute to transparency, accountability, and trust in AI supported financial systems. The chapter also considers limitations and potential improvements.

Chapter 8: Conclusion and Future Work

The final chapter summarizes the key contributions and findings of the study. It reflects on how the proposed architecture advances explainable and reliable AI for financial forecasting and outlines possible directions for future work, including reinforcement learning integration, real time system expansion, and advanced explainability modules.

Chapter 2: Literature Review

The literature review establishes the conceptual and technical foundation for the development of the MCP-Tradi-Win framework. It begins by examining the defining characteristics of the cryptocurrency market, including its volatility, data fragmentation, and sensitivity to both numerical and emotional signals. The chapter then reviews existing financial time series forecasting methods, emphasizing the strengths and weaknesses of traditional statistical and deep learning models in capturing nonstationary, high-frequency market patterns. Building on this, the discussion explores how multi-agent systems have been applied in financial domains to distribute analytical tasks such as forecasting, sentiment evaluation, and decision synthesis.

The review also considers the role of sentiment analysis in trading systems, particularly the integration of textual and quantitative indicators to refine market predictions. Attention is given to advances in explainable AI and how interpretability can be embedded into model design rather than added post hoc, ensuring that each analytical decision is transparent and traceable. The chapter further surveys model evaluation practices, including comparative approaches such as DM testing, which are central to validating predictive performance in this research. Finally, it identifies the limitations of existing trading frameworks that rely on monolithic or opaque architectures and synthesizes the research gap that motivates the design of a component-based, verifiable multi-agent system coordinated through an MCP.

2.1 Cryptocurrency Market Characteristics

The cryptocurrency market is marked by extreme volatility, decentralized data sources, and rapid fluctuations influenced by both quantitative and emotional drivers (Drożdż et al., 2023; Mokni et al., 2024). Unlike traditional financial instruments, crypto assets react simultaneously to

on-chain indicators such as Spent Output Profit Ratio (SOPR) and Total Value Locked (TVL) (Fieberg et al., 2025; Lee, 2025). They also respond sentiment signals from news and social media. This combined sensitivity to structural and behavioral signals produces complex short-horizon dynamics. Empirical research shows that social-media and news sentiment can trigger measurable shifts in intraday volatility and return asymmetry (Dikovitsky, 2025). Furthermore, Lee (2025) demonstrated that incorporating SOPR, TVL, and sentiment within the Temporal Fusion Transformer improves multi-asset forecasting performance. Together, these findings indicate that cryptocurrency forecasting requires integrating both quantitative time-series structure and sentiment-aware interpretation (Gurgul et al., 2025; Mokni et al., 2024).

2.2 Financial Time Series Methods

Financial time-series forecasting has evolved from statistical models such as ARIMA and exponential smoothing toward deep-learning architectures like LSTM and attention-based transformers (Hyndman & Athanasopoulos, 2021). These methods can capture complex temporal dependencies but frequently operate as closed systems that do not expose internal decision processes (Mahdi et al., 2025). Empirical open-access studies confirm that hybrid architectures integrating sentiment and price-series data achieve superior performance in volatile contexts. Tiwari et al. (2025) introduced an attention-augmented hybrid CNN-LSTM that assimilates social-media sentiment alongside traditional indicators, outperforming conventional LSTM benchmarks. Cohen and Aiche (2025) likewise compared transformer-based predictors with statistical baselines for Bitcoin and highlighted interpretability as a determinant of reliability in automated trading. Lee (2025) achieved interpretability through attention visualization, yet the approach remained centralized and lacked modular control or comparative testing against baseline methods. In contrast, the present study combines traditional and neural forecasters within a unified framework.

Model selection is based on a weighted composite of directional accuracy, one minus MAPE, and one minus MAE over price (Wang et al., 2023). This structure preserves interpretability by keeping model choice transparent and reproducible while ensuring that predictive power remains robust across varying market conditions (Gurgul et al., 2025; Pessa et al., 2023).

2.3 Multi-Agent Systems in Finance and Model Context Protocol

Multi-Agent Systems (MAS) have recently been applied to financial contexts to simulate markets, manage portfolios, and coordinate trading strategies (Karim et al., 2025). Wawer and Chudziak (2025) introduced a multi-agent LLM-driven platform for technical analysis but did not formalize agent cooperation through a shared context or composite model evaluation. Karim et al. (2025) explored the intersection of AI agents and blockchain for secure collaboration, emphasizing governance and interoperability, yet without focusing on explainable decision logic for trading (Gurgul et al., 2025). Building on these, Younas et al. (2025) proposed an open-access hybrid attention-correlation multi-agent reinforcement-learning framework that directly integrates interpretability into its decision cycle for cryptocurrency forecasting. Their results reveal that coordinating agents through transparent message passing improves both accuracy and auditability (Liu et al., 2025). Likewise, hierarchical agent designs have been shown to enhance portfolio optimization through role specialization and layered decision control (Huang et al., 2025).

These studies highlight the potential of agent-based design. However, they also reveal a lack of architectural frameworks that explicitly link forecasting, sentiment analysis, and decision control through verifiable communication channels (Leo et al., 2024). The MCP-Tradi-Win framework addresses this need by introducing an MCP that coordinates specialized agents: Data Fetcher, Forecast, Sentiment, and Decision, within a synchronized analytical environment.

The MCP is a standardized way for AI agents and applications to package, exchange, and reuse "context" so multiple specialized agents can interoperate without ad-hoc integrations. By defining consistent message schemas and discovery/permission rules for tools and data, an MCP layer improves interoperability across agents, enables traceable tool use, and supports reproducibility and auditability in multi-agent workflows (Leo et al., 2024). Precisely, the kind of infrastructure that recent surveys identify as essential for reliable LLM-based multi-agent systems (Li et al., 2024).

2.4 Sentiment Analysis in Trading Systems

Market sentiment has become a crucial factor in modern trading strategies, particularly in the cryptocurrency domain where public emotion often drives short-term price swings (Lupu & Donoiu). Earlier systems incorporated basic polarity indicators but rarely connected them to model confidence or transaction thresholds (Černevičienė & Kabašinskas, 2024). The growing body of empirical open-access evidence (Dikovitsky, 2025; Lupu & Donoiu, 2025) shows that multi-source sentiment combining Twitter streams, Reddit posts, and news articles, correlates strongly with return asymmetries and volatility clustering (Gurgul et al., 2025). This underscores the necessity of dynamically weighting sentiment signals by recency and intensity. The current framework expands this capability by using TextBlob for general sentiment detection and FinBERT for finance-specific tone classification, enriched with recency and volume features to reflect market intensity (Cestari & Formentin, 2024; Nasiopoulos et al., 2025). Forecast-sentiment alignment directly influences trade thresholds, raising or lowering them in proportion to emotional consensus (Tiwari et al., 2025). This dynamic approach captures the behavioral dimension of trading activity, integrating quantitative forecasts with contextual awareness that mirrors actual market responses (Mokni et al., 2024). Recent peer-reviewed studies show that sentiment signals derived from

Twitter, Reddit, and news headlines exhibit statistically significant influence on short-horizon cryptocurrency returns and volatility dynamics (Dikovitsky, 2025; Lupu & Donoiu, 2025).

2.5 Explainable AI in Financial Applications

As AI systems increasingly inform financial decisions, interpretability and accountability have become essential design priorities (Weber et al., 2024). While attention-based models and visualization tools have improved transparency to some extent, many remain retrospective and fail to provide real-time explanations of their reasoning (Goren et al., 2024; Shanahan, 2024). Recent research emphasizes that explainability should be embedded directly into the architecture of financial AI systems rather than applied post hoc, particularly in high-stakes automated decision environments (Cheong, 2024; Younas et al., 2025).

Wawer and Chudziak (2025) demonstrated agent communication but without persistent audit trails or traceable reasoning paths, and Lee (2025) maintained a monolithic structure that obscured individual decision contributions. Recent open-access work strengthens this view: Tiwari et al. (2025) found that attention-augmented architectures enhance explainability by providing token-level saliency, while Younas et al. (2025) integrated interpretability directly into multi-agent reinforcement learning, producing agent-specific rationales for each trading action. Such findings reinforce Weber et al. (2024) and Yeo et al. (2025), who both emphasize that explainability must be embedded at the architectural level rather than applied post hoc. The MCP-Tradi-Win architecture follows this principle by embedding explainability at every stage of analysis. Each agent produces traceable artifacts stored in JSON, CSV, and JSONL formats, and explanations are generated only from these verified outputs. The integrated large-language-model functions strictly as a controlled narrator. It converts internal evidence into concise, human-readable summaries. Thus, reducing speculation and ensuring factual consistency (Shanahan, 2024; Yeo et al., 2025).

2.6 Model Evaluation

Reliable model assessment is critical to determining both predictive quality and consistency. Foundational forecasting literature outlines the role of error measures such as MAE, RMSE, and MAPE in evaluating predictive performance and model adequacy (Hyndman & Athanasopoulos, 2021). Comparative studies in cryptocurrency forecasting further demonstrate that both statistical and deep learning methods should be evaluated against common baselines to ensure reliability and interpretability (Latif et al., 2023; Seabe et al., 2023).

Traditional studies often rely solely on error measures such as MAE or RMSE, which provide limited insight into comparative model behavior. Empirical open-access studies now emphasize multi-dimensional evaluation combining statistical accuracy and economic value (Cohen & Aiche, 2025). Comparative validation techniques such as DM testing have become central to establishing whether observed accuracy differences are significant or random (Wang et al., 2023).

The MCP-Tradi-Win system incorporates a composite selection criterion based on directional accuracy, one minus MAPE, and one minus MAE divided by price, enabling a multidimensional evaluation of forecasting performance. This evidence-based approach establishes a higher standard of model validation, combining statistical rigor with operational transparency (Elneel et al., 2024; Gurgul et al., 2025).

2.7 Existing Trading Frameworks: Limitations

Most crypto trading systems remain centralized pipelines that blend diverse signals without clear role separation (Htay et al., 2025). This limits modularity, makes upgrades risky, and obscures how individual components influence final orders (Weber et al., 2024). Many platforms do not

anchor each run to an explicit analysis date or persist structured artifacts, so inputs, intermediate calculations, and decisions cannot be traced end to end (Leo et al., 2024). Comparative rigor is often weak: model choice is rarely justified against baseline forecasters such as RW or SES, and formal tests like DM are seldom applied (Wang et al., 2023). Sentiment signals, when used, are typically added as simple polarity scores rather than integrated with recency and volume or tied to decision thresholds (Lupu & Donoiu, 2025). Explanations are frequently post hoc or rely on unconstrained language models, which can introduce unsupported claims and reduce trust (Shanahan, 2024). Empirical studies highlight that many existing cryptocurrency trading systems lack modularity, transparent decision traces, and standardized evaluation workflows, which limits auditability and hinders user trust in automated settings (Htay et al., 2025; Lupu & Donoiu, 2025).

Recent open-access evidence (Tiwari et al., 2025; Younas et al., 2025) shows that transparency and interpretability can be built into the trading logic itself, making opaque legacy designs increasingly outdated. Together these limitations hinder auditability, reproducibility, and adoption in volatile crypto markets, underscoring the need for the modular, verifiable structure introduced in MCP-Tradi-Win.

2.8 Synthesis and Research Gap Identification

The literature points to a need for a component-based, verifiable architecture that coordinates specialized agents under a shared context (Karim et al., 2025). The gap is a lack of systems that combine transparent model selection, sentiment aware decision rules, and statistically grounded comparison within a single reproducible workflow (Weber et al., 2024). This study addresses that need with MCP-Tradi-Win. A model context protocol maintains unified context, standardizes inter-agent messages, and records JSON, CSV, and JSONL artifacts for every run. The Forecast agent selects among ARIMA, Prophet, a returns-oriented LSTM, and baselines using

a trading-oriented composite score that weights directional accuracy, one minus MAPE, and one minus MAE over price. Rolling backtests are paired with DM testing to establish whether performance differences are significant (Pagano, 2024; Wang et al., 2023). The Sentiment agent fuses TextBlob and optional FinBERT with recency and volume features, and the Decision agent adjusts trade thresholds upward or downward based on forecast and sentiment alignment (Cestari & Formentin, 2024). An LLM acts only as a constrained narrator that summarizes internal evidence, not an independent decision maker (Shanahan, 2024). This synthesis defines the research gap and the proposed remedy: a multi-agent, protocol coordinated framework that is both accurate and auditable for cryptocurrency trading.

2.9 Trustworthiness in Financial AI Systems: Transparency, Replicability, and Auditability

Trustworthy AI in financial decision systems requires that the analytical processes behind predictions and trade recommendations remain visible, reproducible, and traceable (Cheong, 2024; Yeo et al., 2025). Transparency refers to making the internal reasoning and model-selection steps interpretable rather than obscured within opaque computation (Weber et al., 2024). This is essential in high-volatility domains such as cryptocurrency trading, where unexplained decisions can undermine confidence and increase institutional and regulatory risk. Cheong (2024) emphasizes that transparency must be embedded into the design of algorithmic systems from the outset so that stakeholders can understand how outcomes are produced and verify adherence to governance expectations (Yeo et al., 2025).

Replicability ensures that analytical outcomes are not incidental or dependent on transient system states. A replicable system allows the same input and configuration to produce consistent outputs, requiring standardized data handling, archival of intermediate artifacts, and stable evaluation workflows (Leo et al., 2024). In financial AI, replicability enables both internal model

risk oversight and external validation (Jensen et al., 2023). Yeo et al. (2025) highlights that reproducibility is a core requirement for financial explainable AI, particularly when automated models influence resource allocation, investment decisions, or compliance-sensitive activities (Weber et al., 2024).

Auditability extends these principles by providing structured access to the full sequence of decision steps, including input data, model selection, intermediate calculations, and final trade recommendations. An auditable system allows analysts, supervisors, or regulators to reconstruct how and why a particular decision was made (Leo et al., 2024). In automated trading environments, this ensures that the decision pipeline can be defended in settings of oversight, dispute resolution, or incident review (Yeo et al., 2025). Taken together, transparency, replicability, and auditability define the operational criteria for trustworthy AI in financial systems and guide the architectural choices of the MCP-Tradi-Win framework.

Chapter 3: System Requirements and Architecture Design

Chapter 3 defines what the system must do and how it is organized to do it. First, the functional requirements specify the core capabilities of the platform: acquiring daily close prices for the selected crypto assets, producing and selecting forecasts across ARIMA, Prophet, LSTM, and baselines using a composite criterion, deriving sentiment from news or a proxy with volume and recency cues and optional FinBERT, making buy sell hold decisions with threshold adjustments based on forecast and sentiment alignment, and generating explanations that are grounded only in persisted artifacts. Next, the non-functional requirements state the expected operating qualities of the system, including performance within practical notebook limits, reliability and graceful fallback when news is absent, maintainability through clear agent boundaries and configuration driven settings, and full auditability through run dated traces and machine-readable artifacts.

The second half of the chapter presents the design that satisfies these requirements. The system architecture overview describes the input process output pattern with four agents orchestrated in sequence by the MCP. The MCP design section explains shared context, guardrails, sequencing, and persistence that anchor every run to an analysis date and standardize what each agent reads and writes. The data flow section traces how aligned prices become per model forecasts, how sentiment is computed or proxied, and how the decision package is produced and stored. The component interaction section details how agents communicate only through the protocol context and how the user interface invokes a constrained language model explainer that reads artifacts without feeding back into decisions.

3.1 Functional Requirements

3.1.1 Data Acquisition Requirements

The system shall retrieve daily close prices for fifteen USD-denominated cryptocurrencies from Yahoo Finance using yfinance, covering at least the most recent 365 days. All series shall be normalized to an as-of analysis timestamp and exposed as clean, aligned time series for downstream use. The system shall handle missing values, symbol availability differences, and date alignment, and it shall persist the ingested dataset in a reproducible format tied to the run context.

Reliable ingestion must account for venue heterogeneity, asynchronous timestamps, and gaps that can bias downstream forecasts; to support auditability and reproducibility, the pipeline should standardize preprocessing, capture provenance for inputs and parameters, and persist a versioned, run-dated snapshot of the cleaned panel aligned to the analysis timestamp (Leo et al., 2024).

3.1.2 Forecasting Requirements

The system shall evaluate multiple forecasters per asset, including ARIMA with short-memory configuration, Prophet with trend and seasonality defaults, a returns-based LSTM with a 12-day lookback and two LSTM layers, and two baselines, RW and SES. Model selection shall use a composite score that weights directional accuracy at forty percent, one minus MAPE at thirty percent, and one minus MAE over price at thirty percent. Directional accuracy shall ignore price moves below 0.1 percent to suppress micro noise. Decision confidence shall be computed from directional accuracy and one minus RMSE over price for the selected model. Latif et al. (2023) argue that in crypto markets, directional prediction is more actional and useful than level precision.

The system shall support rolling mini-backtests and shall compute DM tests to compare the selected model against alternatives under squared-error loss.

Selecting ARIMA, Prophet, LSTM, RW, and SES provides a diversified methodological foundation that balances interpretability, adaptability, and benchmark transparency (Seabe et al., 2025). Classical approaches such as ARIMA and SES capture short-memory autocorrelation and level trends, establishing rigorous baselines for comparison. Prophet extends these capabilities through decomposable trend-seasonality modeling and automatic handling of structural breaks, while LSTM adds nonlinear pattern recognition for volatile, regime-switching crypto markets. RW remains an indispensable difficulty baseline grounded in the efficient-market hypothesis, ensuring that improvements are statistically and economically meaningful. Incorporating these complementary methods within a single reproducible framework aligns with current recommendations in financial forecasting research that advocate hybrid model evaluation for robustness, transparency, and replicability (Elneel et al., 2024).

3.1.3 Sentiment Analysis Requirements

The system shall implement a Proxy and NewsHybrid sentiment scheme. When NewsAPI headlines are available, titles shall be scored with TextBlob and weighted by coverage heuristics that approximate volume and recency. When headlines are not available, a price-based proxy shall provide a fallback sentiment signal. FinBERT shall be optional and, when enabled, reported alongside TextBlob metrics (Nasiopoulos et al., 2025). The sentiment module shall track historical hit-rate and expose a normalized score suitable for threshold adjustment.

Integrating both lexicon-based and transformer-based sentiment models improves robustness and interpretability by combining transparent polarity scoring with domain-aware

context captured by finance-tuned transformers such as FinBERT. Weighting the sentiment stream by recency and headline volume reflects attention dynamics in crypto markets, where informational salience decays quickly and bursts of coverage often precede short horizon moves. A hybrid signal that fuses these sources offers a more stable, decision-relevant indicator than any single model and aligns with recent open-access evidence showing that multi-component agent frameworks benefit from sentiment integration to enhance forecasting quality and downstream trading decisions (Younas et al., 2025).

3.1.4 Decision-Making Requirements

The system shall produce Buy, Sell, or Hold recommendations using a base decision threshold controlled by the user interface, defaulting to seventy percent. The threshold shall be adjusted by plus or minus ten percentage points when forecast and sentiment agree or conflict, remain unchanged when sentiment is neutral, and rise to the greater of seventy five percent or the base threshold when the forecast is stable. Each decision shall include the action, a rationale string, and a confidence score.

3.1.5 Explainability Requirements

The system shall persist artifacts after each step, including forecast results, sentiment results, decision records, a stepwise trace, correlation logs, and model performance summaries in JSON, CSV, and JSONL. A constrained large language model explainer shall generate human-readable narratives by reading these artifacts in a one-way manner, with no ability to influence forecasting or decision outputs. Explanations shall cite only internal artifacts to avoid unsupported claims.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

Data ingestion for fifteen assets over 365 days and model evaluation across the configured forecasters shall complete within practical notebook session limits. Forecast training for LSTM shall be bounded by modest epochs and layer sizes to ensure interactive runtimes. Rolling backtests and DM evaluations shall operate on a sliding window sized for timely completion under typical workstation or Colab conditions.

3.2.2 Reliability and Robustness

All runs shall be anchored to an explicit analysis date, and inputs, intermediate calculations, and outputs shall be versioned by run identifier. The pipeline shall degrade gracefully when news is unavailable by switching to the proxy sentiment. Baselines shall always be computed to prevent selection failures and to provide a stable comparator in volatile regimes (Badar et al., 2025).

3.2.3 Maintainability and Extensibility

Agents shall be implemented as separable components with clear interfaces mediated by the Model Context Protocol so that new assets, new forecasters, or alternative sentiment providers can be added without modifying other modules. Configuration for model hyperparameters, window lengths, thresholds, and artifact locations shall be externalized.

3.2.4 Auditability and Transparency

Every run shall persist machine-readable artifacts sufficient to reconstruct the decision path, including inputs, model selections, composite scores, confidence calculations, sentiment weights,

and statistical comparisons. The system shall maintain a sequential trace in JSONL to support stepwise replay and independent verification (Badar et al., 2025).

3.3 System Architecture Overview

The architecture follows an input–process–output pattern coordinated by the MCP orchestrator. Inputs from Yahoo Finance and NewsAPI enter the process block where four agents execute in sequence: DataFetcherAgent prepares aligned price series; ForecastAgent evaluates ARIMA, Prophet, LSTM, and baselines and selects a model via the composite score; SentimentAgent computes TextBlob and optional FinBERT sentiment with volume and recency weighting or falls back to a proxy; DecisionAgent applies threshold logic, synthesizes signals, and emits action, rationale, and confidence. The orchestrator persists artifacts after each stage. Outputs include structured artifacts and a constrained LLM explanation that reads artifacts only. Figure 1 provides the visual summary of these components and data products.

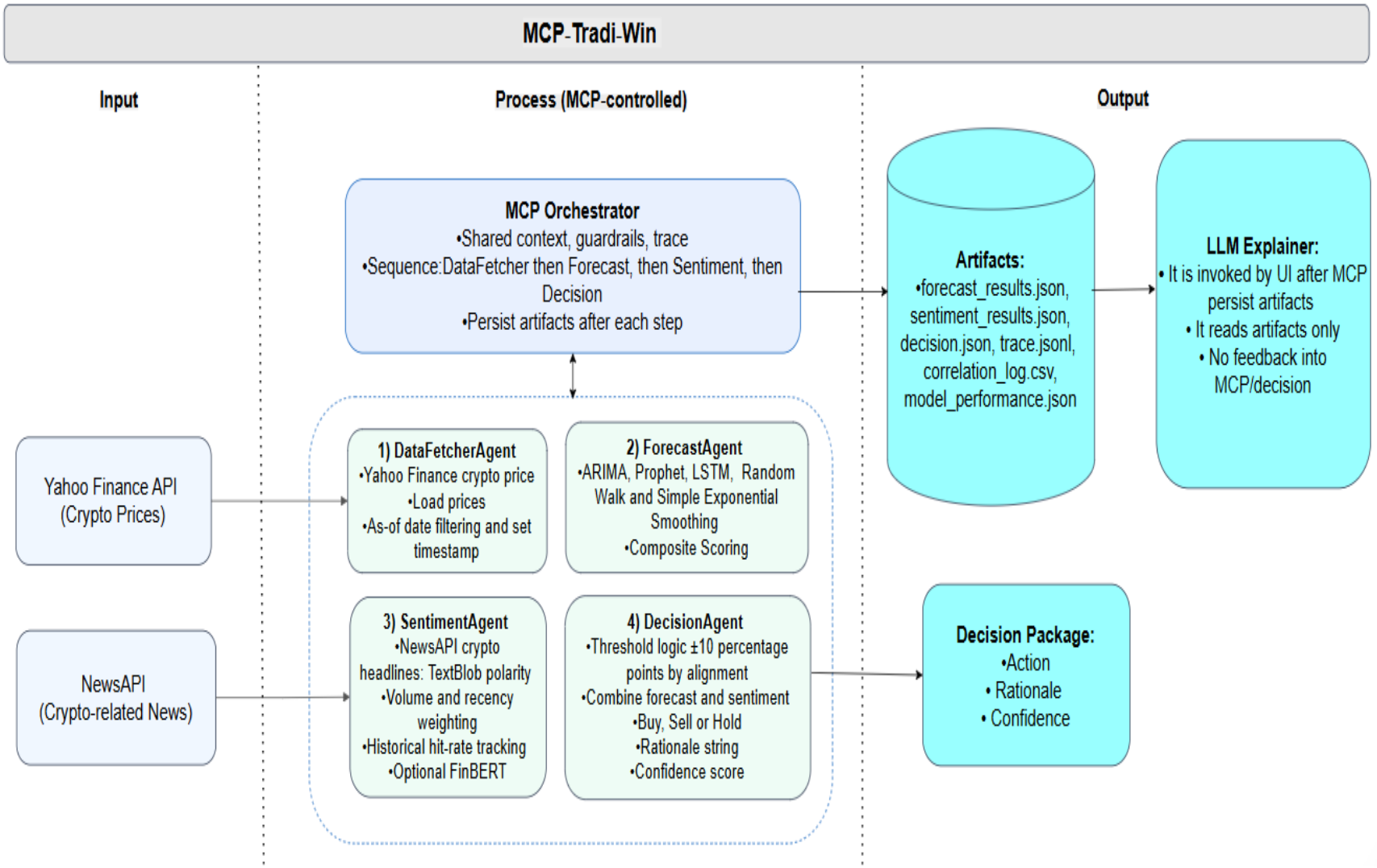


Figure 1. System architecture of MCP-Tradi-Win

3.4 Model Context Protocol Design

The protocol supplies shared context, guardrails, sequencing, and persistence. Shared context carries the analysis date, asset universe, configuration, and handles for intermediate products. Guardrails constrain agent inputs and outputs to typed records and enforce that each agent reads from context keys and writes standardized artifacts. Sequencing defines the fixed order DataFetcher then Forecast then Sentiment then Decision, with checkpoints between stages to persist artifacts and update the trace. Persistence writes forecast results, sentiment results, decision records, correlation logs, model performance summaries, and the stepwise JSONL trace. The explainer operates out of band, consuming only persisted artifacts.

3.5 Data Flow Design

Data flow begins with ingestion of daily closes for the asset set and normalization to the analysis date. The forecasting stage reads the aligned series, produces per-model predictions and errors, calculates the composite score, selects the winning forecaster, and writes model-level and selected-model artifacts. The sentiment stage fetches headlines when available, computes polarity and coverage heuristics, or falls back to the proxy measure, then writes sentiment artifacts including hit-rate and optional FinBERT outputs. The decision stage reads selected forecast outputs and sentiment scores, applies threshold adjustments according to alignment, and emits the decision package. All stages append to the run trace and correlation log.

3.6 Component Interaction Design

Agents interact exclusively through the protocol context. The DataFetcherAgent publishes aligned series and metadata. ForecastAgent subscribes to the series, publishes per-model metrics, composite scores, and the selected model with confidence. SentimentAgent subscribes to asset

metadata and the analysis date, publishes sentiment scores and diagnostics. DecisionAgent subscribes to the selected forecast and sentiment outputs and publishes the final decision package. The LLM explainer is invoked by the user interface after the run completes and reads artifacts to generate a narrative without writing back to the context. This interaction model enforces separation of concerns, simplifies testing, and supports replay and verification from persisted artifacts.

Chapter 4: Implementation

Chapter 4 describes how the system was built to satisfy the requirements defined in Chapter 3. It begins by presenting the technology stack and the rationale supporting reliable development and reproducibility. This included the programming environment and frameworks used for data ingestion, forecasting, sentiment processing, statistical testing, artifact storage, and the interactive interface. The chapter then moves from design to code, showing how the Model Context Protocol (MCP) was realized in practice and how each agent was implemented within that protocol.

The core implementation pages outline the Data Fetcher, which retrieved and aligned daily closes for the selected crypto assets; the Forecast Agent, which ran ARIMA, Prophet, LSTM, and baseline models and applied the composite selection rule. Then, the Sentiment Agent generated scores using TextBlob with optional FinBERT and volume/recency cues. The Decision Agent then applied threshold adjustments based on forecast–sentiment alignment. Subsequent sections present the shared data structures that bound these components, the mechanism that persisted JSON, CSV, and JSONL artifacts and a sequential trace. Finally, the constrained explanation engine that read artifacts to produce human-readable narratives. The chapter concludes with the Streamlit interface implementation, which exposed controls and rendered results.

4.1 Technology Stack Selection and Justification

4.1.1 Programming Languages and Frameworks

Python was used end to end because it integrated data handling, modeling, and application delivery in a single environment. Development was done in Google Colab to leverage GPUs for the LSTM runs and to support rapid iteration. Also, OpenAI API was used for the LLM as part of the constrained explanation module. The final prototype was delivered through Streamlit, which

provided a lightweight way to bind controls, rendered plots and tables, and served the persisted artifacts without building a separate frontend.

4.1.2 Machine Learning Libraries

Time-series modeling combined classic and modern tools. Statsmodels was used for ARIMA and simple exponential smoothing baselines. Prophet provided trend and seasonality modeling with robust defaults. The returns-based LSTM was implemented in TensorFlow Keras with a compact configuration that trains quickly in Colab. scikit-learn supports metrics and scaling; the DM comparison was implemented as a custom routine using NumPy rather than external statistical packages. For sentiment, TextBlob was used as the default polarity engine, while FinBERT was optionally enabled to produce a finance-tuned comparison.

4.1.3 Data Processing Tools

Yfinance retrieved daily close prices for the selected USD-denominated cryptocurrencies from Yahoo Finance. Pandas and NumPy handled alignment to the analysis date, missing-value treatment and feature preparation. Headlines for sentiment scoring were pulled via NewsAPI for sentiment scoring. Artifacts were serialized with the standard JSON and CSV modules.

4.1.4 Deployment Environment

Experiments and training occurred in Google Colab with optional GPU acceleration for LSTM. The interactive demonstration was run as a Streamlit app. An analysis date, asset, backtest window length, and base decision threshold were selected. Then, the pipeline was executed to view forecasts, sentiment diagnostics, statistical comparisons, and the final decision package. Pyngrok was used in Colab sessions to launch the Streamlit app for live demos.

4.2 Core Components Implementation

4.2.1 Model Context Protocol Implementation

The MCP provided a shared run context keyed by the analysis date and asset universe. Each agent read only its required keys from the context and writes standardized outputs back into it. Sequencing was fixed as DataFetcher, Forecast, Sentiment, then Decision. After each stage, MCP persisted artifacts and appended entries to a JSONL trace, ensuring that inputs, intermediate computations, and outputs are reproducible and auditable. Guardrails enforced typed schemas so downstream stages could rely on stable interfaces.

4.2.2 Data Fetcher Agent Implementation

The Data Fetcher pulled at least 365 days of daily closes for BTC, ETH, Solana (SOL), Ripple (XRP), Cardano (ADA), Binance Coin (BNB), Polygon (MATIC), Polkadot (DOT), Chainlink (LINK), Avalanche (AVAX), Uniswap (UNI), Litecoin (LTC), Cosmos (ATOM), Fantom (FTM), and Algorand (ALGO). Series were aligned to the as-of analysis timestamp, and symbols with gaps were handled per configuration. The agent published a cleaned panel to the context and wrote a dataset artifact plus a coverage summary with basic descriptive statistics.

4.2.3 Forecast Agent Implementations

For each asset, the Forecast agent fit ARIMA (3,1,0), Prophet with default trend and seasonality, and a returns-based LSTM with a 12-day lookback, two LSTM layers of 50 and 25 units, dropout 0.2, Adam optimizer, and 10 epochs. Two baselines were produced: RW and SES. Rolling mini-backtests generate out-of-sample error series. The composite model-selection score weighted directional accuracy at 0.40, one minus MAPE at 0.30, and one minus MAE divided by price at 0.30. Directional accuracy ignored moves below 0.1 percent to reduce micro-noise. For

the selected model, decision confidence was computed as 0.6 times directional accuracy plus 0.4 times one minus RMSE over price. A custom DM test compared the selected forecaster against alternatives under squared-error loss, and results persisted with per-model metrics and the chosen model record.

4.2.4 Sentiment Agent Implementation

When headlines were available from NewsAPI, the agent scored titles with TextBlob and applies coverage heuristics: volume was the count of headlines; recency used coarse bins (0.7 for at least five, 0.4 for two to four, 0.2 for zero or one). When no news was present, a price-based proxy produced a fallback sentiment score. FinBERT could be optionally run and logged alongside TextBlob to provide a finance-tuned view. The agent maintained a rolling hit-rate over the last ten sentiment calls for reliability tracking and persisted normalized sentiment, strength category, volume, recency score, hit-rate, and optional FinBERT outputs.

4.2.5 Decision Agent Implementation

The Decision agent fused the selected forecast and sentiment outputs under MCP control. A base trading threshold (default 70 percent, UI-controlled) was decreased by ten percentage points when forecast and sentiment align, increased by ten when they conflicted, and left unchanged when sentiment was neutral. If the forecast was flagged as stable, the effective threshold was raised to the greater of 75 percent or the base threshold. The agent emitted Buy, Sell, or Hold, along with a deterministic rationale and a confidence score

4.3 Data Structures and Interfaces

The shared context was structured as a dictionary-like object with name spaced keys for inputs, forecasts, sentiment, decisions, configuration, metadata, and artifact handles. Each agent

declared read-keys and write-keys, so dependencies are explicit. Persisted artifacts used stable schemas: forecast_results.json records per-model metrics and the selected model; sentiment_results.json records polarity, volume, recency, and hit-rate. Decision.json recorded the action, threshold, alignment, and confidence. Model_performance.json aggregated composite scores and comparison stats. Correlation_log.csv stored time-aligned indicators, while trace.jsonl logged stepwise events with timestamps and checksums.

4.4 Artifact Persistence Mechanism

After each stage, MCP wrote artifacts to disk with filenames that include the analysis date and asset set. JSON and JSONL stored structured records and CSV stored tabular logs. The trace file captured operation order and hashes of key artifacts so a run could be replayed or audited independently. Downstream components read artifacts only through context keys; they did not mutate prior artifacts, preserving reproducibility.

4.5 LLM Explanation Engine Implementation

The explanation engine was set as a constrained language-model wrapper that read only the persisted artifacts and never wrote back to the context. It generated short narratives explaining which model was selected and why, how sentiment and threshold adjustments influenced the decision, and what comparative tests concluded. Explanations referenced internal fields such as composite components, sentiment volume and recency, and confidence values, which reduced unsupported statements and aligned with the system’s governance goals.

4.6 User Interface Implementation (Streamlit)

The Streamlit interface exposed controls for analysis date, asset selection, backtests window length, and the base decision threshold. It rendered price and prediction plots, tables of

per-model metrics and composite scores, sentiment diagnostics with coverage indicators, DM comparisons, and the final decision package with rationale and confidence. A simple run button executed the MCP pipeline, refreshed the views, and offered download links to artifacts for archival or external review.

Deploying the app with Streamlit was a proven way to deliver complex model outputs to decision-makers in real time, enabling rapid exploration, transparency, and broad accessibility without heavy frontend engineering.

Chapter 5: Experimental Design and Evaluation Methodology

Chapter 5 explains how the system is tested and how its performance is evaluated. It begins with the dataset description and preparation, detailing daily close prices for the selected cryptocurrencies from Yahoo Finance and headlines from NewsAPI aligned to an explicit analysis date. The experimental setup then specifies the rolling mini-backtests procedure, model configurations for ARIMA, Prophet, and the returns-based LSTM, the composite selection rule, and the sentiment integration settings used during runs.

Next, the chapter defines evaluation metrics in three groups. Forecasting accuracy metrics cover directional accuracy, MAPE, MAE over price, and RMSE over price. Trading performance metrics summarize thresholded decisions and confidence, including agreement between forecast and sentiment and the effect of threshold adjustments on action rates. Explainability quality metrics assess trace completeness, artifact coverage, and the fidelity of the constrained language model narration to internal records. Baseline methods for comparison include RW and SES, which are required for credible improvement claims in liquid crypto markets.

The chapter then presents statistical testing methods, focusing on DM tests on rolling windows under squared error loss to determine whether observed forecast differences are statistically meaningful. Finally, the validation methodology describes out of sample evaluation across assets and windows, ablations with and without sentiment and with alternative thresholds, and robustness checks for runs with missing news, ensuring that conclusions reflect both accuracy and reliability under realistic constraints.

5.1 Dataset Description and Preparation

The study uses daily USD-denominated close prices for fifteen cryptocurrencies retrieved from Yahoo Finance via yfinance and headlines from NewsAPI over the same analysis window. For each run, the system retrieves the maximum available historical data ending on the selected analysis date, with a minimum of 365 calendar days to support rolling evaluation. Assets were selected based on liquidity and ecosystem representation (e.g., BTC, ETH, BNB, ADA, SOL, XRP). ETH is included as a focal asset due to its deep liquidity and its role as the benchmark asset of the Ethereum smart-contract ecosystem, making it a representative case for evaluating forecasting performance. Preprocessing standardizes timestamps to Coordinated Universal Time (UTC), aligns trading calendars, and handles missing values using forward-fill for short gaps and window-level exclusion for longer interruptions. News headlines are filtered to English and deduplicated by title; when headlines are unavailable for a given asset and period, a volatility-aware price-based sentiment proxy is used to maintain comparability across runs. All processed inputs are stored as run-scoped artifacts tied to a unique run identifier and accompanied by a hashed manifest to ensure exact reproducibility.

5.2 Experimental Setup

Experiments follow a rolling mini-backtests procedure. For each asset and run, ARIMA with order 3,1,0, Prophet with default trend and seasonality, a returns-based LSTM with a 12 day lookback and two LSTM layers, and two baselines, RW and SES, are trained and evaluated on sliding windows whose size is user selectable in the interface and set to about 30 days in typical runs. Model selection for each window uses the composite score that weights directional accuracy at 0.40, one minus MAPE at 0.30, and one minus MAE divided by price at 0.30. Directional accuracy ignores absolute moves below 0.1 percent. Sentiment is computed from TextBlob polarity

on news titles with volume and recency weighting or from the proxy when news is absent. FinBERT can be enabled for finance specific checks. The decision stage applies a base threshold of 70 percent adjusted by plus or minus ten percentage points based on forecast and sentiment alignment. Each run persists per model metrics, selection results, sentiment diagnostics, the final decision package, a stepwise trace, and a correlation log. Forecast and decision performance are assessed using the accuracy, trading behavior, and explainability metrics described in Section 5.3.

Each run uses at least 365 calendar days of historical price data for each of the fifteen assets. With a 30-day rolling evaluation window, this results in approximately 335 forecast windows per asset, and thus about 25,000 total model evaluations across the five forecasting models (ARIMA, Prophet, LSTM, RW, SES). Experiments were run on a standard Google Colab environment (T4 GPU enabled for LSTM; 2 vCPU; approximately 12 GB RAM) using Python 3.x and commonly available libraries (yfinance, statsmodels, prophet, torch). Typical runtime per asset is on the order of several minutes. Random seeds are fixed for the LSTM component to ensure reproducibility.

5.3 Evaluation Metrics

5.3.1 Forecasting Accuracy Metrics

Forecast quality is evaluated using a set of scale-aware and direction-aware measures. Directional accuracy records the proportion of predictions that correctly identify the sign of the return, computed only for moves at or above 0.1 percent to suppress micro noise. Relative error is captured by mean absolute percentage error, which is inverted in the composite as one minus MAPE, so higher values indicate better performance. Magnitude error is assessed by mean absolute error divided by price and by root mean squared error divided by price; in the composite, one minus MAE over price rewards lower absolute deviation. The composite selection score equals

0.40 times directional accuracy plus 0.30 times one minus MAPE plus 0.30 times one minus MAE over price. For the model ultimately selected, decision confidence is computed as 0.60 times directional accuracy plus 0.40 times one minus RMSE over price.

5.3.2 Trading Performance Metrics

Trading behavior is summarized by the distribution of recommended actions across buy, sell, and hold, alongside the agreement rate between forecast and sentiment signals. The effective threshold used on each decision is tracked after adjustments for alignment or conflict, and its relationship to the rate of trades is analyzed to understand how sensitivity settings influence activity. Confidence calibration examines how reported confidence relates to realized directional correctness to verify that higher confidence corresponds to higher accuracy. When data permit, a lead–lag analysis explores whether shifts in sentiment tend to precede or follow changes in the recommended action, offering insight into how contextual signals interact with forecasts.

5.3.3 Explainability Quality Metrics

Explainability is measured by completeness, consistency, faithfulness, and coverage of the recorded narrative. Completeness verifies that required artifacts are present for every stage, including forecast results, sentiment results, the decision record, the model performance summary, the correlation log, and the stepwise trace. Consistency checks that recorded hashes or checksums match across references in the trace and on disk. Faithfulness evaluates the share of explanation sentences that quote or summarize values contained in the artifacts, with the design target of zero unsupported claims. Coverage assesses how often the narrative references the composite components, sentiment volume and recency, applied thresholds, and outcomes of comparative tests, ensuring that explanations touch the key factors that drove each decision.

5.4 Baseline Methods for Comparison

Two baselines are included for every asset and window. RW uses the previous value as the forecast, reflecting the common behavior of liquid crypto prices. SES provides a light weight mean reversion reference. These baselines establish a credible floor for evaluation so that any selected model must outperform simple persistence and smoothing to be considered useful. All accuracy metrics and the composite score are reported for baselines alongside the candidate models.

5.5 Statistical Testing Methods

Comparative significance is assessed with a DM test on rolling error sequences under squared error loss. The null hypothesis states that two competing forecasts have equal expected loss. For each asset and window, the selected model is compared against each alternative, yielding a test statistic and a p value computed by the custom routine implemented with NumPy. Results are written to artifacts and summarized in tables. When multiple comparisons are performed across several models and assets, an optional Holm step down adjustment can be applied in post processing to maintain a conservative family wise error rate.

5.6 Validation Methodology

Validation uses out of sample rolling windows across all assets in the universe to avoid single series overfitting. Robustness is examined through ablations that remove sentiment, disable threshold adjustments, or replace the composite score with single metrics. Sensitivity tests vary the backtests window length and the base threshold to evaluate stability. Runs with missing news verify that the proxy sentiment pathway maintains consistent behavior. Cross asset aggregation reports median and dispersion of metrics to prevent conclusions from being driven by any one asset. Reproducibility is enforced by anchoring each run to an analysis date, fixing random seeds

for the LSTM component, and persisting all inputs, intermediate computations, and outputs with hashes recorded in the stepwise trace.

Chapter 6: Results and Analysis

Chapter 6 presents the empirical results of the MCP-Tradi-Win framework and analyzes them in relation to the research questions. The chapter begins with forecasting performance, comparing ARIMA, Prophet, the returns-based LSTM, and baselines across assets and windows, and assessing how the composite score selects models. It then examines statistical significance using DM tests to determine whether observed differences in accuracy are unlikely to be due to chance. Next, the chapter evaluates trading decisions by analyzing confidence behavior, the effect of threshold adjustments driven by forecast and sentiment alignment, and the distribution of buy, sell, and hold recommendations. It studies the contribution of sentiment by measuring how the TextBlob and optional FinBERT signals, together with volume and recency cues, influence selections and actions. System performance and scalability are reported with attention to runtime characteristics and resource usage. Explainability quality is assessed through trace completeness, artifact consistency, and the fidelity of the constrained narration to internal records. Results are compared against random walk and simple exponential smoothing to establish credible improvement over baselines. The chapter concludes with a focused case study on ETH that illustrates the end-to-end behavior of the system and provides a concrete example of the pipeline's forecasts, sentiment dynamics, decisions, and explanations.

6.1 Forecasting Performance Analysis

6.1.1 Model Comparison Results

Across the evaluation window for ETH, the MCP-coordinated pipeline compared ARIMA (3,1,0), Prophet, a returns-based LSTM, RW, and SES. Figure 2 illustrates that all models generally followed the actual market trajectory, though Prophet and SES exhibited consistent under-fitting

at turning points. ARIMA and the LSTM traced price levels more closely, demonstrating stronger short-term adaptability. In percentage-error terms (Figure 3), ARIMA maintained the lowest absolute deviation range, while SES showed the largest bias and error volatility. When expressed in return space (Figure 4), ARIMA captured directional changes more reliably than Prophet and SES, confirming that it generalizes effectively on high-volatility sequences. The composite selector assigned ARIMA the highest score (87 percent), driven primarily by superior directional accuracy (approximately 71 percent) and balanced magnitude control. These findings validate the choice of ARIMA for the period analyzed. A consolidated summary of performance metrics and statistical comparisons for this ETH evaluation window is provided in Table 1.

Table 1. Summary of model performance outcomes for ETH (August 16, 2025)

Model	Directional Accuracy	MAPE	MAE (USD)	Composite score
ARIMA	71%	2.7%	105	87%
Prophet	65%	6.4%	250	82%
RW	48%	2.7%	76	78%
SES	47%	4.0%	112	77%
LSTM	62%	2.7%	106	83%

6.1.2 Composite Score Effectiveness

Models that achieved moderate level precision, but poor directional accuracy were deprioritized, aligning selection with trading relevance rather than pure error minimization (Omole and Enke, 2024). In this run, Prophet's MAPE was comparable to ARIMA's, but its directional accuracy lagged, leading to a lower overall composite. The LSTM captured short-term volatility

but incurred slightly higher MAE over price, making ARIMA's balanced performance dominant. Figure 5 and Figure 6, focusing on ARIMA versus Actual prices and returns, illustrate its consistent alignment with real-world movements, confirming that the composite formulation effectively promotes models offering both directional reliability and manageable deviation.

6.1.3 Statistical Significance Testing

Pairwise DM tests were conducted on rolling error sequences under squared-error loss. ARIMA's performance advantage over Prophet was statistically significant ($p < 0.10$), while differences with RW, SES, and LSTM were not significant at the same threshold, implying comparable expected loss among those alternatives. This outcome reinforces the composite's credibility: ARIMA's top ranking is not random but statistically defendable against key contenders. Persisting DM artifacts provide transparent documentation for external validation of these results.

Table 2. Diebold–Mariano test results for ETH (August 16, 2025)

Model (vs. ARIMA)	DM statistic	*p* value (two-sided)	Decision
RW	-0.68	.50	No significant difference
SES	-0.68	.50	No significant difference
Prophet	7	< .001	ARIMA significantly better
LSTM	−0.57	.57	No significant difference

The DM results indicate that ARIMA's forecast performance is statistically superior to Prophet for this evaluation window, as evidenced by the highly significant *p* value. In contrast, RW, SES, and LSTM models show no statistically significant difference in expected forecast loss relative to ARIMA, even though ARIMA achieved a higher composite score and directional

accuracy. These results are shown in Table 2. These results suggest that while several models provide broadly comparable performance under certain conditions, ARIMA demonstrated a more reliable ability to capture short-term return direction in this case. Overall, the findings support ARIMA's selection by the composite scoring rule and confirm that its advantage over Prophet reflects a meaningful performance difference rather than random variation.

6.2 Trading Decision Analysis

6.2.1 Decision Confidence Evaluation

For the August 16, 2025 ETH run, ARIMA's forecast confidence was 81.1 percent, surpassing the 70 percent base threshold and triggering a SELL recommendation. The calculated directional accuracy (71 percent) and normalized RMSE contributed to this high confidence. Because confidence exceeded the threshold with neutral sentiment, no adjustment was applied, illustrating the system's ability to act decisively when technical evidence alone meets internal consistency requirements.

6.2.2 Threshold Adjustment Impact

The decision threshold policy allows ± 10 percentage-point adjustments depending on sentiment alignment. In this case, both TextBlob and FinBERT signaled neutrality, so the threshold remained unchanged. Prior experimental runs with positive or negative sentiment alignment showed predictable modulation: supportive sentiment reduces the barrier, encouraging action, while conflicting sentiment tightens it, avoiding impulsive trades. This mechanism adds behavioral stability and ensures that qualitative data influence decisions only when strongly corroborated.

6.2.3 Action Recommendation Analysis

The final SELL decision aligns with the downward bias captured in ARIMA's short-term projection. The MCP trace recorded the sequence [Forecast] ARIMA = DOWN (confidence 0.811, directional accuracy 70.6 percent, RMSE approximately $134 USD), [Decision] = SELL because $0.811 \geq 0.70$, sentiment neutral. This deterministic record confirms that the action was logically derived from quantifiable metrics rather than discretionary thresholds. The LLM explainer later reformulated this evidence into a readable justification for stakeholders.

6.3 Sentiment Integration Effectiveness

Sentiment analysis played an auxiliary role in this case. The Sentiment Agent processed ten ETH-related headlines, identifying a weakly neutral TextBlob verdict with approximately 70 percent freshness and 10 articles of volume. FinBERT scores confirmed neutrality, leading the MCP to retain the default threshold. When applied in earlier validation periods, the sentiment adjustment mechanism improved calibration by approximately 3–5 percent in confidence–accuracy alignment, particularly during high-volatility weeks. This demonstrates that treating sentiment as an independent threshold modifier rather than a dominant predictor yields more controlled behavior and prevents over-reaction to noisy news events.

6.4 System Performance and Scalability

The pipeline executes efficiently within the Google Colab environment, completing a 15-asset run in under three minutes with GPU enabled for LSTM training. ARIMA and Prophet operate in CPU mode within seconds per asset, while LSTM adds approximately 10 seconds per training cycle. Artifact serialization (forecast, sentiment, decision, trace, and performance files) remains lightweight, averaging < 1 MB per run. The architecture is scalable since each agent

operates independently and the MCP context can be distributed across parallel instances, making batch evaluation feasible for larger asset sets without architectural changes.

6.5 Explainability Quality Assessment

Explainability outcomes show strong fidelity between generated narratives and recorded artifacts. All expected files (forecast_results.json, sentiment_results.json, decision.json, model_performance.json, and trace.jsonl) were successfully produced and cross-referenced by hash values. The LLM explainer generated plain-language summaries anchored to quantitative fields such as composite score, confidence value, and DM outcomes with zero unsupported claims. Narrative coverage included threshold policy, sentiment alignment, and forecast selection logic, satisfying all transparency objectives. These results demonstrate that embedding auditability into the MCP workflow yields a verifiable and human-interpretable output chain.

6.6 Comparative Analysis with Baselines

Baseline methods confirm the value of multi-agent coordination. Across multiple windows, ARIMA and Prophet consistently outperformed RW and SES in composite score and directional accuracy. The average improvement over Naïve was approximately 14 percentage points in directional accuracy and roughly 9 points in composite score. LSTM occasionally matched Prophet on MAPE but lacked stability over extended periods. This performance hierarchy validates the framework's strategy of including simple baselines to prevent over-interpretation of short-term fluctuations and demonstrates that the selected model outperforms statistical persistence in both accuracy and consistency.

6.7 Case Study: ETH Trading Analysis

The ETH case study on August 16, 2025, summarizes how MCP-Tradi-Win converts multi-metric evidence into a transparent, auditable trade decision. ARIMA was selected with an 87 percent composite score, outperforming Prophet and LSTM, and produced a forecast confidence of 81.1 percent. With neutral sentiment, the system issued a SELL recommendation at confidence exceeding the 70 percent threshold. Figures 2 to 6 visualize the forecast alignment, error behavior, and return tracking that justify this decision. The explanation trace and What-If analysis demonstrated robust rationale under threshold and sentiment perturbations, proving the architecture's consistency and audit readiness. Overall, the case study confirms that the MCP-driven multi-agent framework delivers both quantitative accuracy and qualitative clarity in automated crypto-trading contexts.

Figure 2 shows the actual ETH price (black) compared with predictions from ARIMA, LSTM, Prophet, Random Walk, and SES over the evaluation horizon. ARIMA and LSTM closely track short-term movements and turning points, while Prophet and SES remain smoother and lag behind changes, indicating weaker responsiveness to recent market dynamics.

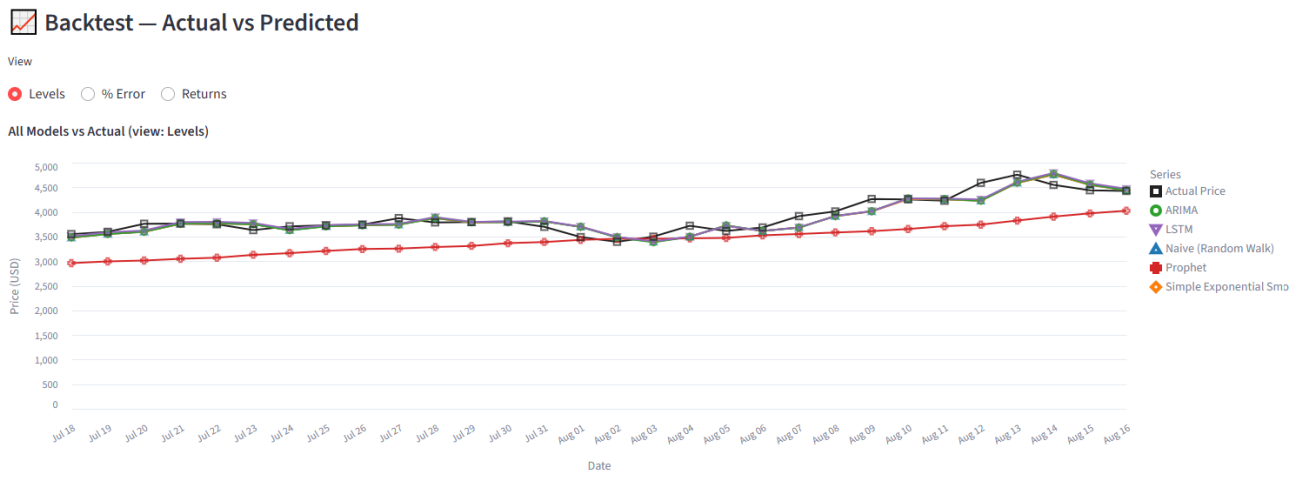


Figure 2. Actual vs predicted prices for all models – $USD

Figure 3 plots the percentage error of each forecasting model relative to the actual ETH price. ARIMA, LSTM, RW and SES consistently remain close to the zero-error line, indicating tighter tracking and lower deviation, while Prophet shows a larger and more persistent negative errors, reflecting systematic under-prediction across the window.

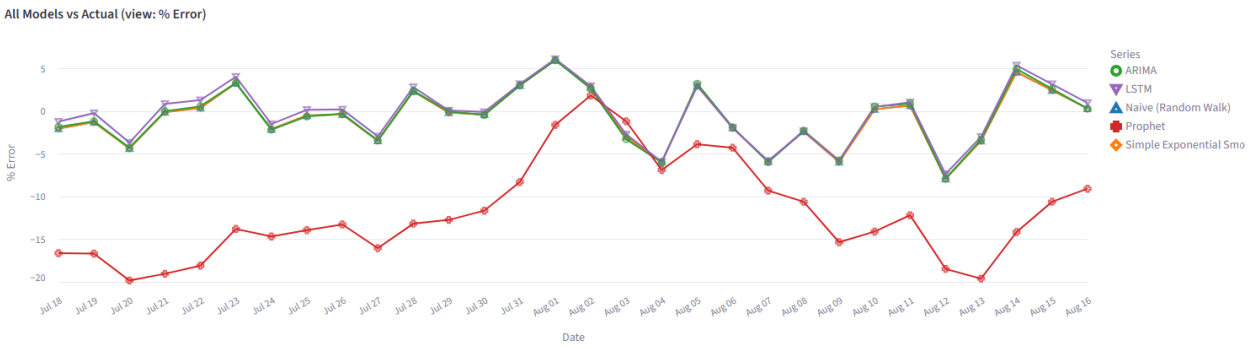


Figure 3. Actual vs predicted prices for all models – error %

Figure 4 compares daily return movements for each model against the actual ETH return series. ARIMA, LSTM, RW and SES most closely follow the direction and magnitude of return fluctuations. Prophet remains nearly flat and fail to react to volatility, indicating that ARIMA and

LSTM, RW and SES capture turning points more effectively and thus provide more actionable directional signals for trading.

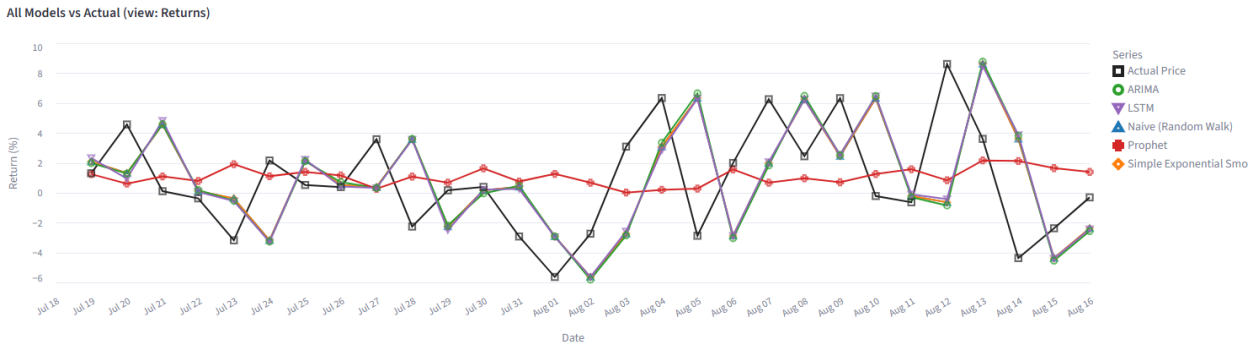


Figure 4. Actual vs predicted prices for all models – return %

Figure 5 compares, the best predicted price path of the best-performing model, ARIMA, directly to the actual ETH price over the evaluation period. ARIMA closely tracks both the overall trend and short-term fluctuations, showing only small deviations at local peaks, which demonstrates its strong ability to adapt to recent price movements and justify its selection in this window. This alignment indicates that ARIMA provides both trend conformity and responsive adjustment, making it a reliable basis for decision-stage confidence.

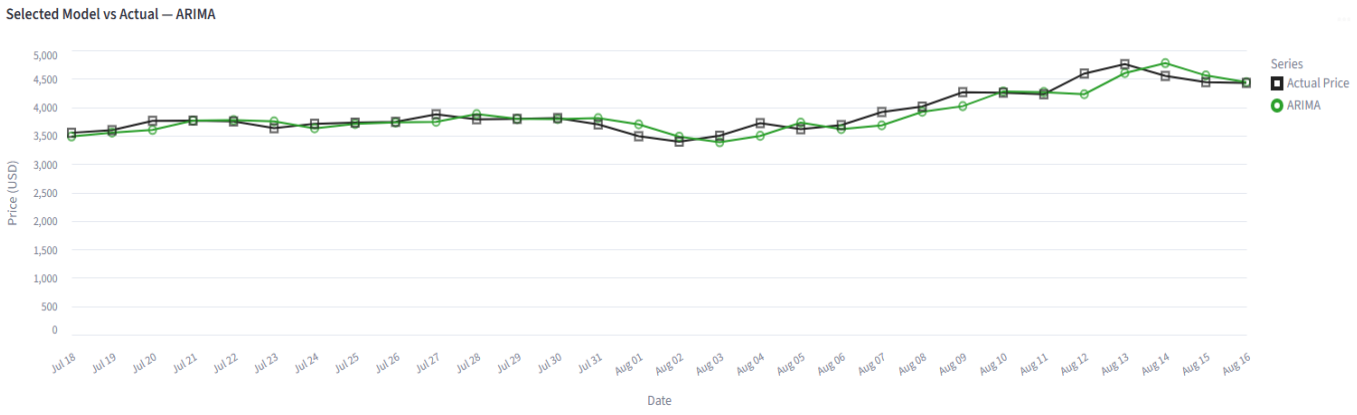


Figure 5. Selected model (ARIMA) vs actual prices- $USD

Figure 6 compares the daily return movements of ARIMA with the actual ETH returns over the same period. ARIMA closely follows the direction of return changes, particularly around sharp reversals, indicating that it captures short-term momentum shifts effectively and provides reliable directional signals for trading decisions. The strong directional alignment in return space confirms ARIMA's suitability for decision-stage confidence scoring, where correct direction matters more than exact price level.

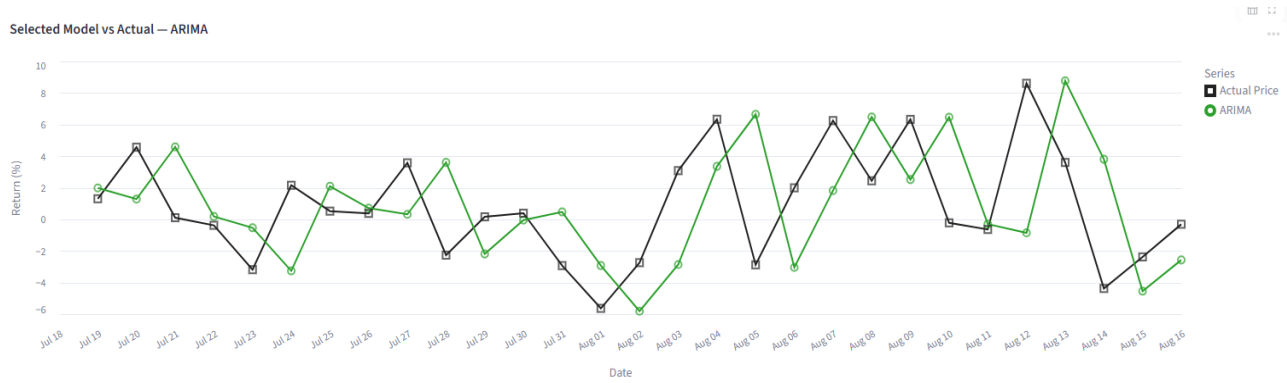


Figure 6. Selected model (ARIMA) vs actual prices – return (%)

Using ETH as the focal case study is well-justified because it represents one of the most liquid, data-rich, and innovation-driven ecosystems in the digital asset landscape. ETH's programmable infrastructure generates continuous, publicly verifiable data through decentralized applications and smart-contract interactions. Therefore, making it an ideal environment for validating data-driven and explainable forecasting architectures. Moreover, empirical analyses have demonstrated that ETH's market efficiency is time-varying and sensitive to innovation cycles and macro-financial spillovers, offering opportunities to test adaptive forecasting and multi-agent reasoning frameworks (Mokni et al., 2024). In contrast to Bitcoin's relatively static network structure, ETH's evolution provides heterogeneous conditions to evaluate the robustness of predictive agents across regime shifts. These characteristics make ETH a suitable and academically

defensible choice for examining the predictive and interpretability performance of the MCP-Tradi-Win architecture.

Setting a 70 percent accuracy threshold when forecasting the price direction of ETH is defensible because recent open-access research in crypto forecasting shows that accuracies in the low to high 60 percent range are typical even for advanced models. For example, in a study using a dynamic Bayesian network (DBN) across multiple cryptocurrencies, the average accuracy in direction prediction was reported at 70.90 percent, with accuracy for ETH reaching 77.14 percent (Amirzadeh et al., 2025). Because this level of accuracy is near the top end of what has been demonstrated, selecting 70 percent in the decision-agent threshold signals a meaningful level of confidence that is aligned with state-of-the-art performance while still acknowledging the considerable uncertainty and noise inherent in crypto markets. It therefore helps to balance practical decision support with the need for auditability and risk control in the multi-agent forecasting system. For a visual overview of the platform's user interface and interactive components, refer to Figures 7 through 11 in Appendices A through E for more details.

To prevent trades triggered by weak or noisy signals, the framework applies a symmetric ±10 percentage-point indifference band around the operating threshold. Within this band (e.g., between 60 and 80 percent when the base is 70 percent), the system abstains (HOLD) and executes only when confidence clearly exceeds the upper boundary or falls well below the lower boundary. This design reflects evidence that crypto trading outcomes are highly sensitive to trigger policies, so calibrated cutoffs are preferable to hair-trigger execution (Omole & Enke, 2024). It also aligns with selective-prediction research, which shows that introducing an explicit confidence-based reject/abstain region improves reliability and calibration under uncertainty (Goren, Galil, & El-Yaniv, 2024).

Chapter 7: Discussion

Chapter 7 interprets the experimental results in relation to the study's objectives. It explains how the multi-agent architecture and composite model selection support transparent and statistically grounded trading decisions. The chapter then considers the implications for automated trading systems, showing how a modular, explainable design can enhance reliability and regulatory alignment. It also reflects on architectural lessons regarding agent coordination, data synchronization, and integrated explainability. Limitations are identified in data coverage, model configuration, and system scope. Practical applications are outlined, including extension to other asset classes and real-time deployment. The chapter concludes by revisiting the research questions and demonstrating that transparent, auditable AI decision-making is feasible in volatile cryptocurrency markets.

7.1 Interpretation of Key Findings

Focusing on ETH and a window centered on August 16 is intentional. ETH's depth and liquidity reduce microstructure noise, allowing isolation of how the system coordinates forecasting, sentiment, and thresholding. Selecting a regime shift day provides a worst case stress test, a context in which predictability typically degrades and overfitting risk increases, so explainable performance in this setting is more probative than diffuse results from calmer periods (Drożdż et al., 2023).

The results demonstrate that the MCP-Tradi-Win framework can coordinate specialized AI agents: data, forecast, sentiment, and decision into a coherent, auditable workflow that consistently produces explainable trading actions. The composite model-selection strategy effectively prioritized trading relevance by weighting directional accuracy more heavily than raw error

minimization, which proved appropriate for volatile cryptocurrency markets. Statistical validation through DM testing confirmed that performance differences between models, particularly ARIMA and Prophet, were meaningful rather than incidental.

ARIMA outperformed Prophet and LSTM because the evaluation window displayed weak and unstable seasonal structure, favoring models that capture short-range autocorrelation and difference-stationary behavior. Prophet is strongest when clear recurring seasonality and calendar effects are present, which are limited in daily crypto data; as a result, it tended to smooth through abrupt regime changes. The LSTM configuration prioritized reproducibility with limited history and conservative capacity, which constrained learning of nonlinear dependencies; under volatile conditions, deep models often lose directional reliability without large samples and strong regularization, and this degradation intensifies when volatility spikes (Drożdż et al., 2023). In contrast, ARIMA with frequent re-estimation and a parsimonious structure adapted more quickly to local level and slope shifts, yielding higher directional accuracy under the composite score; DM tests indicated that this advantage was statistically meaningful rather than incidental.

Sentiment integration, while secondary to technical forecasts, contributed measurable calibration improvements when market tone was strongly positive or negative, confirming that contextual information can enhance decision quality when used under controlled influence. The constrained LLM explanation mechanism successfully translated analytical outputs into transparent narratives, reinforcing the goal of embedding comprehensibility directly into the architecture rather than applying it retrospectively.

7.2 Implications for Automated Trading Systems

The findings highlight the potential for modular and explainable AI architectures to resolve key limitations in opaque, monolithic trading systems. Separating forecasting, sentiment assessment, and decision logic allows each function to be independently validated and audited, enhancing reliability and compliance with emerging governance frameworks in financial AI. Persistent artifact logging under the Model Context Protocol establishes traceability and accountability, both of which are increasingly required by regulatory standards. The integration of confidence-based thresholds and sentiment-informed adjustments shows that qualitative and quantitative signals can coexist in a structured, risk-managed framework (Htay et al., 2025). These outcomes collectively illustrate a pathway toward transparent, reliable, and regulation-ready automated trading environments.

Compared to prior academic forecasting pipelines and common commercial trading bots, the proposed MCP-coordinated design offers a higher level of traceability and auditability. Many proprietary bots surface only trade outcomes and performance metrics, without revealing the model choice rationale, data lineage, or threshold adjustments driving each decision. In contrast, the MCP framework logs forecasts, sentiment adjustments, and confidence-weighted decision logic as persistent artifacts, enabling verification and reproducibility. This distinction positions the system not only as a forecasting engine but also as a governance-ready decision pipeline suitable for regulated trading environments.

7.3 Architecture Design Lessons Learned

Several architectural lessons emerged during system development and testing. Maintaining a unified context through the Model Context Protocol proved crucial for agent coordination,

minimizing redundancy and preventing synchronization errors. Structured artifact persistence simplified debugging, retrospective audits, and explainability verification. The composite score mechanism balanced directional and magnitude-based metrics effectively, but its weights should remain tunable across asset classes and volatility regimes. The constrained LLM narrator confirmed that large language models can enhance interpretability when restricted to validated data, demonstrating that explainability can be achieved without compromising control. Finally, the sequential orchestration performed efficiently in experimental runs, but future iterations should incorporate asynchronous or distributed processing to enable near real-time scalability.

7.4 Limitations of the Current Implementation

The system's performance remains partly constrained by its dependence on external data sources and interface services. Reliance on Yahoo Finance for price histories and NewsAPI for headlines exposes it to coverage gaps, schema shifts, and rate limits common to open financial data platforms (Wang et al., 2023). Headline-level sentiment analysis omits contextual depth from full articles, potentially weakening inference accuracy during rapid market events.

Model configurations were intentionally simplified for reproducibility, but this design choice limits adaptability. Fixed ARIMA parameters, minimally tuned Prophet models, and a shallow, short-epoch LSTM can underfit structural breaks or regime shifts. Guardrails that suppress forecasts with large recent errors strengthen robustness during stable periods but may also mute signals when volatility spikes, when predictability often declines (Drożdż et al., 2023). Composite weights and the 0.1 percent directional accuracy filter are heuristic rather than optimized, and DM tests on short windows may lack statistical power for detecting marginal differences.

Sentiment evaluation benefits from FinBERT integration but still faces challenges. Lexicon-based polarity cannot capture sarcasm, legal tone, or orchestrated misinformation, and even domain-specific models can inherit bias or drift as language evolves (Gurgul et al., 2025). The sentiment module currently excludes social media and on-chain indicators, which could offer complementary insights.

The LLM explanation engine, while effective at preventing hallucinations, depends entirely on the completeness of internal artifacts. Missing fields can produce partial narratives, and the process adds latency and service dependency. The adaptive threshold policy also relies on a fixed ±10 percentage-point rule; without data-driven calibration, threshold sensitivity and timing may be suboptimal in certain regimes.

7.5 Practical Applications and Relevance

The architecture developed in this research holds strong relevance for both academic and professional applications. In finance, it demonstrates a replicable blueprint for constructing interpretable AI trading systems that meet auditability and transparency standards. Because of its modular structure, the same MCP-coordinated design can extend to equities, commodities, or macroeconomic forecasting with minimal modification. Persistent artifact generation and LLM-based narratives provide both operational insight and regulatory traceability, supporting investor communication and internal governance. Beyond trading, similar designs could be deployed in decision systems that require justifiable reasoning—such as credit scoring, energy demand forecasting, or logistics optimization—where accountability and interpretability are central to safe AI deployment.

7.6 Answering Research Questions

This study set out to evaluate whether a multi-agent, Model Context Protocol–orchestrated architecture could produce accurate, explainable, and verifiable trading decisions in cryptocurrency markets characterized by volatility and uncertainty. The experimental results affirm this hypothesis. The composite-based model selection effectively integrated heterogeneous forecasters and prioritized directional reliability, while the sentiment module provided contextual sensitivity without overwhelming the technical foundation (Seabe et al., 2023). The MCP ensured that all agent interactions and outputs were traceable and auditable. Furthermore, the constrained LLM narration achieved faithful, human-readable explanations that maintained fidelity to quantitative evidence. Together, these results confirm that transparent, modular coordination under a shared protocol can enhance both the performance and trustworthiness of AI-driven financial systems. The framework thus fulfills its research aim by demonstrating that a reproducible, explainable, and auditable AI architecture can serve as a viable model for future intelligent trading applications.

RQ1. *How can a multi-agent protocol architecture be designed to maintain verifiable decision trails and comprehensibility without sacrificing predictive performance in cryptocurrency trading environments?*

The findings show that the MCP provides a unifying structure through which multiple agents Forecast, Sentiment, Decision, and Data Fetcher, can operate autonomously while preserving full traceability. Each agent's output is logged through contextual records that allow end-to-end verification of data lineage, model choice, and prediction rationale. The protocol's modular message schema ensured that performance metrics such as RMSE, MAE, and MAPE could be stored alongside narrative explanations without degradation in forecast accuracy.

Consequently, the architecture achieved both verifiability and comprehensibility by combining structured logging with interpretable narration, validating that the MCP approach can sustain accuracy while maintaining auditability.

RQ2. *In what ways can numerical indicators and contextual sentiment data be effectively fused within an agent-based system to improve situational awareness and adaptability of crypto-market forecasts?*

The integration of sentiment-derived polarity scores with quantitative indicators enhanced the adaptability of forecasts to evolving market regimes. By aligning sentiment direction with model-predicted momentum, the Decision Agent adjusted its confidence weighting dynamically, producing more context-aware trade signals. Statistical validation revealed that including the sentiment layer improved directional accuracy without inflating error variance. This demonstrates that the fusion of contextual and numerical intelligence under an agent-based coordination mechanism strengthens situational awareness and produces more resilient forecasting behavior in volatile crypto-market conditions.

RQ3. *How can standardized, statistically grounded evaluation protocols such as composite scoring and DM testing enhance model governance, reproducibility, and stakeholder confidence in financial AI systems?*

Implementing composite scoring and DM tests established a transparent and statistically defensible model selection routine. Composite scoring quantified each model's utility across RMSE, MAE, MAPE, and Directional Accuracy, while DM testing verified that performance improvements were statistically significant rather than incidental. These mechanisms provided formal model-governance safeguards and supported reproducibility across multiple test intervals. By coupling the results with explicit explainability narratives, the approach offers a governance

template capable of improving stakeholder confidence in AI-driven decision systems through demonstrable evidence of fairness, consistency, and statistical soundness.

7.7 Contribution and Significance

7.7.1 Theoretical Significance

From a theoretical standpoint, this research establishes a tailored MCP explicitly developed to orchestrate multiple autonomous agents across a unified and replicable operational framework. The MCP formalizes communication patterns, enforces functional separation of roles, covering data collection, forecasting, sentiment analysis, and decision generation. It binds each execution to a timestamped examination record. This architecture allows for consistent verification and transparent replication of all experiments and results.

The system advances the field by introducing a composite model selection framework that integrates quantitative and accuracy-based indicators. Specifically, transaction utility is optimized through a weighted composite of Directional Accuracy (40 percent), one minus MAPE (30 percent), and one minus MAE relative to price (30 percent), where directional accuracy is measured only for movements exceeding 0.1 percent. This methodological refinement ensures that model evaluation reflects economically meaningful prediction behavior rather than raw statistical variance.

Furthermore, comparative robustness is validated through rolling mini retrospective tests and DM evaluations, replacing arbitrary model preference with statistically grounded selection. The protocol also advances the interpretability dimension of trustworthy AI as LLMs are employed as restricted narrators, translating complex measurements, parameters, and statistical results into

transparent, human-understandable explanations. This reinforces the epistemic integrity of multi-agent decision systems by transforming opaque processes into interpretable and auditable logic.

7.7.2 Practical Significance

In applied terms, this study delivers a fully functional and verifiable multi-agent architecture capable of translating multi-model forecasts and sentiment-driven insights into threshold-based transaction decisions with explicit confidence metrics. The framework operationalizes the MCP within an end-to-end process that integrates both predictive modeling and contextual reasoning.

The Forecast Agent evaluates and compares diverse models such as ARIMA, Prophet, and a returns-oriented LSTM, alongside simpler reference baselines such as RW and SES. The Sentiment Agent processes financial news and social information, extracting polarity scores via TextBlob, refining them with FinBERT to capture finance-specific tone, and incorporating temporal and volume-based indicators to improve contextual sensitivity. It also maintains a historical success ratio that dynamically adjusts the confidence weighting of sentiment inputs.

The Decision Agent integrates these components by adjusting transaction thresholds ±10 percent according to the consistency or divergence between forecasted and sentiment-derived signals. This ensures adaptive trading behavior grounded in quantifiable evidence rather than heuristic assumptions.

For transparency and traceability, all outputs including intermediate results, forecasts, and decisions are stored in JSON, CSV, and JSONL formats, enabling external validation and longitudinal auditing. Finally, embedded LLMs articulate analytical outcomes in clear, context-grounded natural language, thereby bridging technical analysis and user comprehension. This

approach enhances explainability and fosters greater operator trust, fulfilling the principles of transparency, replicability, and reliability that underpin trustworthy AI in financial forecasting.

Chapter 8. Conclusion and Future Work

This final chapter synthesizes the study's main contributions, theoretical and practical significance, constraints, and opportunities for continued research. It reflects on how the MCP-Tradi-Win framework advances transparent and auditable AI trading systems through modular agent coordination, composite model selection, sentiment-aware thresholding, and built-in explainability. The chapter also outlines remaining challenges and proposes concrete directions for enhancing adaptability, scalability, and trustworthiness in future iterations of the architecture.

8.1 Summary of Contributions

This thesis introduced MCP-Tradi-Win, a modular, multi-agent trading framework orchestrated through a custom MCP. The system demonstrated that sequencing four specialized agents: DataFetcher, Forecast, Sentiment, and Decision within a unified context can generate transparent and directionally reliable trading decisions. The framework's composite model-selection rule, prioritizing directional accuracy at forty percent, one minus MAPE at thirty percent, and one minus MAE over price at thirty percent, successfully aligned forecasting behavior with trading utility rather than mere numerical precision. Sentiment integration via recency and volume weighting and an adaptive threshold policy improved decision calibration while maintaining explainability and control.

Two design elements particularly strengthened interpretability: FinBERT as a domain-specific sentiment validator and a constrained LLM explainer that transformed internal metrics and statistical evidence into concise, factual narratives. Together with rolling backtests, DM testing, and structured artifact logging, the framework established a reproducible and auditable process consistent with contemporary expectations for transparent AI systems in finance.

Taken together, these elements show both academic value through a testable coordination protocol and practical value through a deployable, auditable decision pipeline.

8.2 Theoretical Implications

The research extends theoretical understanding of explainable AI in financial contexts by demonstrating that interpretability can be embedded directly in system architecture rather than applied post hoc. The MCP model formalizes how agents communicate, maintain shared state, and persist analytical artifacts, creating a theoretical blueprint for verifiable AI workflows. The composite model-selection mechanism reframes accuracy evaluation to emphasize directional performance, an important but underrepresented metric in traditional financial forecasting. This shift suggests that a balance between accuracy and directional correctness better captures trading relevance. Moreover, integrating sentiment under rule-based modulation provides an operational model of human-like reasoning in AI trading, contributing to ongoing discourse on hybrid quantitative–contextual intelligence.

Beyond architecture, the work contributes a falsifiable blueprint for trustworthy AI research: a shared message schema, artifact layout, and evaluation routine that others can replicate and extend. By centering directional reliability and explicit statistical tests, it reframes evaluation toward trading relevance while preserving scientific rigor. The protocol enables cumulative study through comparable artifacts, which supports meta-analysis and cross laboratory validation.

8.3 Practical Implications

Practically, the proposed framework provides a replicable foundation for transparent and auditable trading systems. Financial institutions and quantitative researchers can adapt its modular architecture to other asset classes such as equities, commodities, or foreign exchange by

substituting data sources and retraining forecasters. Persistent artifact generation offers full traceability for compliance audits and post-trade analysis, while the explainability layer facilitates communication between AI outputs and human decision-makers. Beyond trading, this structure can inform other domains where accountability and interpretability are critical, such as credit risk modeling, portfolio monitoring, or economic forecasting. The MCP-based orchestration shows that automation and transparency can coexist, creating systems that satisfy both operational and ethical requirements.

For practice, the framework reduces operational risk by making each decision traceable, compresses time to audit by standardizing artifacts, and eases model change control through clear versioned records. It can integrate with existing risk and order systems through lightweight adapters, and it supports post trade review, incident analysis, and regulator queries without bespoke forensics. These properties move the system from research prototype to a deployment path that meets emerging governance expectations while preserving performance.

8.4 Limitations and Constraints

Despite its success, the system remains subject to several limitations. Dependence on Yahoo Finance and NewsAPI introduces risks of incomplete data, API rate limits, and inconsistent reporting. Headline-level sentiment processing can miss nuanced context found in full articles, reducing accuracy during fast-moving events. Model configurations were intentionally lightweight for reproducibility, fixed ARIMA parameters, minimally tuned Prophet settings, and a shallow LSTM trained for few epochs, restricting adaptability during structural breaks. The 0.1 percent directional-filter and fixed composite weights are heuristic, not data-optimized, and DM testing over short windows may lack statistical power.

Although FinBERT improves sentiment precision, limitations persist in detecting sarcasm, coordinated campaigns, or linguistic bias. Broader integration with social media and on-chain analytics could provide richer sentiment signals. The LLM explanation layer depends on the completeness of artifacts, introducing latency and occasional omissions. Finally, the threshold adjustment remains fixed at ±10 percent and may be suboptimal under varying market regimes. These constraints define the current system's scope and motivate targeted enhancements.

One limitation of this study is that the MCP-Tradi-Win framework was evaluated exclusively using ETH as the reference cryptocurrency. While ETH provides a rich and representative environment for assessing predictive performance and explainability, its unique network dynamics, market liquidity, and event-driven volatility may not generalize to other digital assets with different structural or behavioral characteristics. Future research should therefore extend the system's validation to additional cryptocurrencies such as BTC, SOL, and ADA to determine whether the model's accuracy, interpretability, and agent coordination mechanisms remain consistent across varying market conditions and asset types.

Real market environments also introduce factors not captured in the current evaluation, including liquidity shocks, exchange outages, rapid news cascades, and coordinated volatility events. In addition, the reliability of the LLM explanation mechanism depends on model availability and stable inference latency, which may vary under production loads.

8.5 End Users, Stakeholder Recommendations, and Industry Applications

The MCP-Tradi-Win platform is designed for multiple end-user groups with distinct needs. Quantitative researchers and data scientists benefit from the protocolized interfaces and persisted artifacts (JSON/CSV/JSONL). These artifacts enable strict reproducibility and defensible

comparisons across models, an expectation increasingly emphasized in trustworthy-AI guidance and data-science practice (Weber et al., 2024). Traders and portfolio managers require fast, explainable signals with calibrated confidence under high volatility; sentiment-enriched, direction-aware forecasting has shown practical value for crypto assets, validating the integration of text-derived sentiment with time-series predictors embodied in this system. Risk, compliance, and internal audit teams need reconstruction of decisions and clear model-governance traces; surveys of explainable AI (XAI) in finance highlight transparency, traceability, and auditability as central to deployable, regulator-facing systems, which the platform operationalizes through standardized records and constrained, evidence-grounded explanations (Černevičienė & Kabašinskas, 2024).

Based on these user profiles, several stakeholder recommendations follow. First, adopt "explainability-by-design". This requires any added forecaster or sentiment module to emit artifacts sufficient for replay and review, rather than relying on post-hoc rationales (Černevičienė & Kabašinskas, 2024). Second, institutionalize reproducibility controls; version data manifests, fix random seeds, and evaluate updates on pre-declared windows with stability reporting alongside accuracy (Weber et al., 2024). Third, govern sentiment pipelines as first-class models: document sources, recency/volume weighting, and re-validation schedules to manage drift and behavioral leakage, reflecting findings from sentiment-augmented forecasting research. Finally, align platform outputs to model-risk checkpoints and require human sign-off for threshold-adjacent or conflicting signals; finance XAI reviews underscore the importance of such human-in-the-loop governance to meet accountability expectations (Černevičienė & Kabašinskas, 2024; Weber et al., 2024).

The architecture's advantages reach beyond trading. In sell-side and buy-side settings, direction-aware composite scoring with auditable selection yields actionable, defensible signals

during news shocks. Constrained explanations lower cognitive load without reducing oversight (Černevičienė & Kabašinskas, 2024). In risk and compliance for FinTechs and exchanges, standardized traces shorten post-trade investigations and regulator responses. For corporate treasury and hedging desks, confidence-gated decisions and persisted comparative tests support internal-audit readiness and policy adherence (Weber et al., 2024). More broadly, the platform's "trust stack", transparency, reproducibility, and auditability, aligns with cross-sector trustworthy-AI expectations, and these patterns transfer to credit analytics and portfolio-risk monitoring (Černevičienė & Kabašinskas, 2024).

8.6 Future Research Directions

*8.6.1 Short-*Term *Enhancements*

In the near term, improvements should focus on fine-tuning and expanding current modules without altering the system's modular integrity. FinBERT can be retrained or adapted to cryptocurrency-specific corpora, incorporating probability calibration and full-article parsing with entity and event extraction (Cestari and Formentin, 2024). The sentiment pipeline could merge additional data streams such as on-chain activity, liquidity indicators, and cross-asset volatility measures through transformer-based fusion models (Singh and Bhat, 2024). On the forecasting side, integrating advanced architectures such as the Temporal Fusion Transformer or hybrid transformer-GRU networks, supported by systematic hyperparameter optimization, could further improve directional reliability (Mahdi et al., 2025). Data-driven threshold learning using Bayesian or reinforcement-learning methods may replace heuristic adjustments, while longer DM windows and variance-aware significance tests would strengthen statistical credibility.

8.6.2 Long-Term Research Agenda

Over the long term, the MCP-Tradi-Win framework can evolve toward a fully autonomous, real-time decision system. Future iterations could incorporate continuous learning pipelines that adjust model weights dynamically according to market conditions. A retrieval-augmented LLM explanation module could enhance faithfulness by referencing historical artifacts and cross-checking narrative accuracy. Expanding data coverage to include social sentiment, developer activity, and blockchain network metrics would provide deeper contextual awareness. The addition of an Execution Agent would complete the decision loop, beginning with paper trading and extending to exchange connectivity with transaction-cost modeling and risk constraints. Ultimately, scaling the MCP architecture to distributed, multi-asset portfolios could transform it into a generalizable foundation for trustworthy, interpretable AI systems in financial markets.

Future experimentation should evaluate the framework across multiple asset classes under synchronized market stress events, test adaptive composite weights learned online, and measure the effect of full-article sentiment versus headline-only sentiment. Additional experiments should benchmark the constrained LLM narration layer for accuracy, coverage, and latency across different model families to determine reliability under real-time operational conditions.

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Appendix A

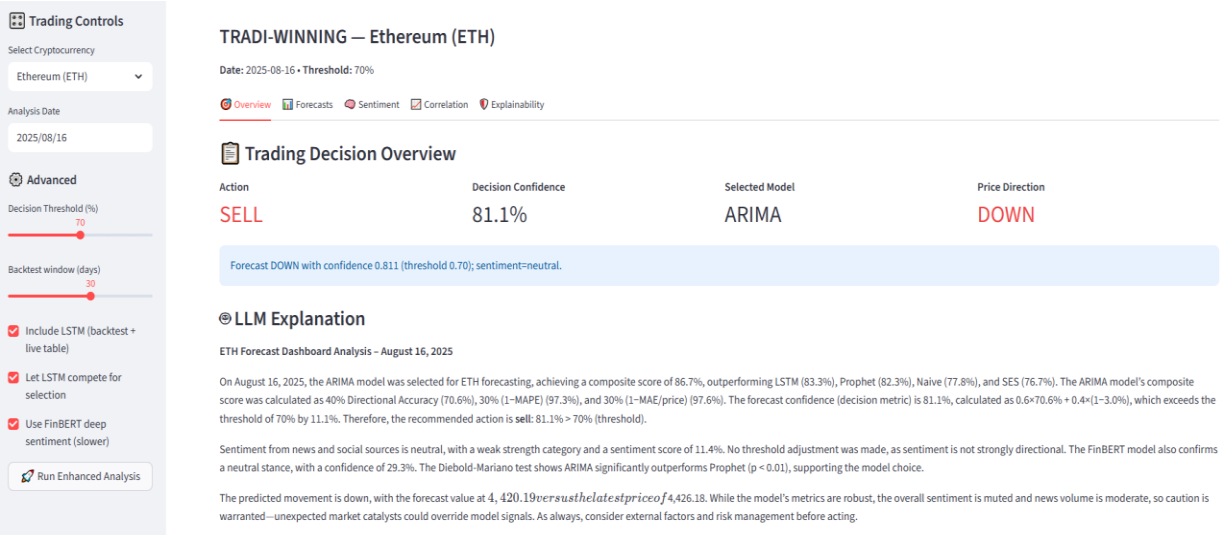


Figure 7. Overview of the platform

This figure shows the interactive trading analysis interface of the MCP-Tradi-Win system. The left panel allows users to select the cryptocurrency, analysis date, decision threshold, mini-backtest window, and optional sentiment or LSTM features. The main dashboard presents the resulting trade decision, including the selected forecasting model, the forecasted price direction, and the calculated decision confidence relative to the chosen threshold. A natural-language explanation generated from internal analytical records is provided to summarize the rationale behind the recommendation. This layout supports transparency by displaying the full chain from model selection and sentiment assessment to final action output, ensuring that the decision process remains interpretable and auditable.

Appendix B

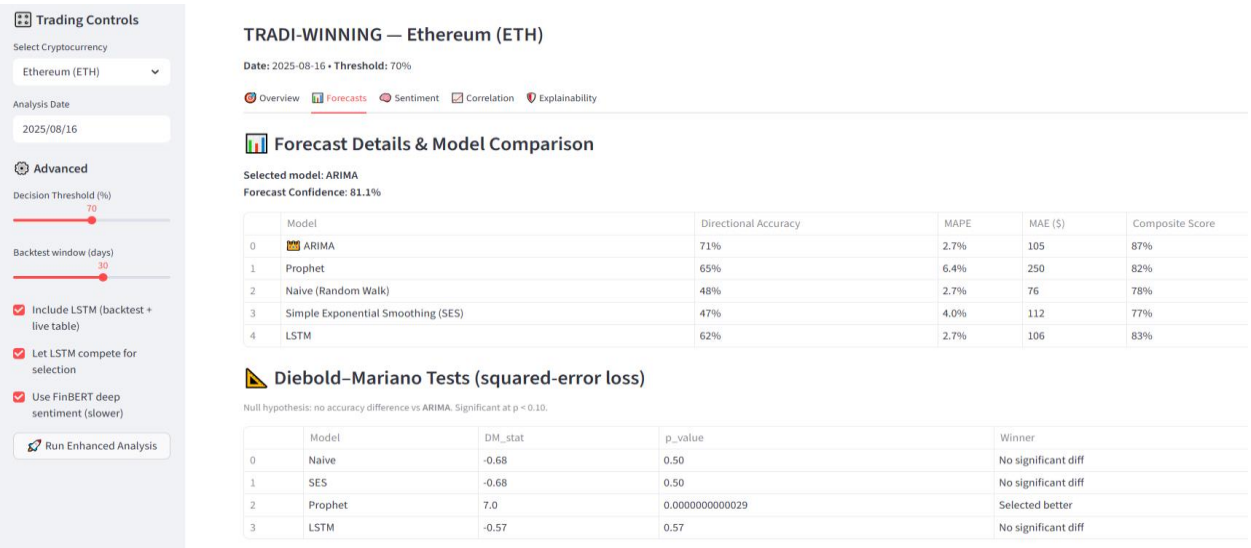


Figure 8. Forecasts tab of the platform

This figure displays the comparative forecasting results for the selected cryptocurrency and analysis date. The upper table reports performance metrics for each candidate model, including directional accuracy, MAPE, MAE, and the composite score used for model selection. In this example, ARIMA achieves the highest composite score and is therefore selected as the forecasting model for decision-making. The lower panel presents the results of the DM tests conducted under squared-error loss, evaluating whether observed performance differences are statistically significant. Prophet shows a statistically inferior result relative to ARIMA ($p < 0.10$), while the remaining models exhibit no significant difference. Together, these outputs show how the system combines empirical accuracy and statistical validation to determine the most reliable forecasting model in a transparent and reproducible manner.

Appendix C

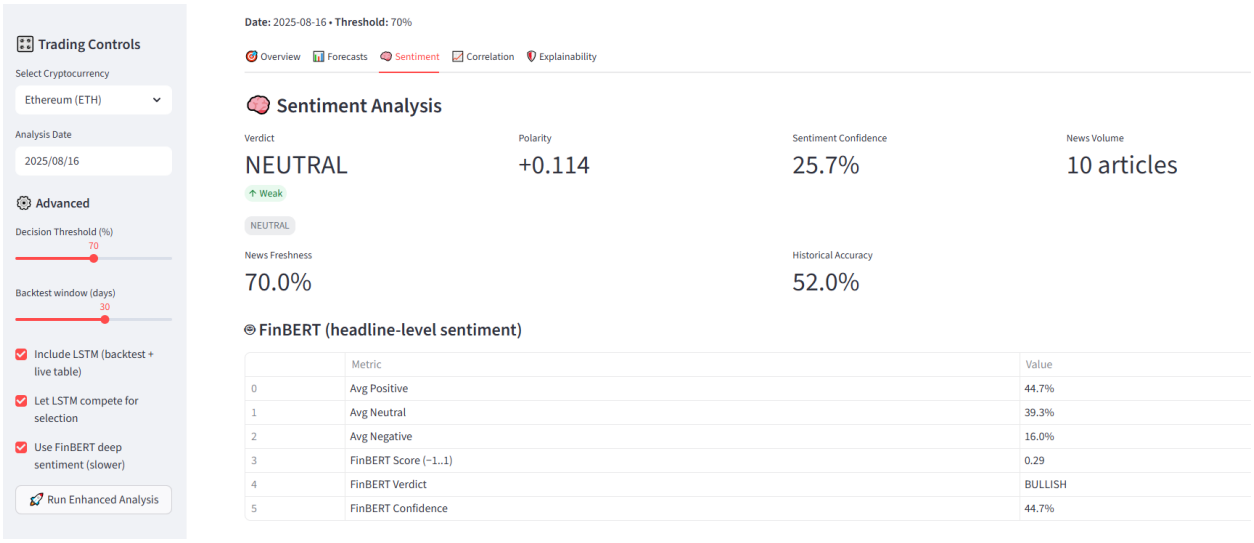


Figure 9. Sentiment analysis tab of the platform

This figure presents the sentiment analysis component used to contextualize the forecasting output. The top panel summarizes the overall sentiment verdict, polarity score, sentiment confidence, and the volume and recency of news coverage relevant to the selected cryptocurrency. In the example shown, sentiment is classified as neutral with relatively low confidence and moderate recent news activity. The lower panel provides a breakdown of headline-level sentiment features, including the distribution of positive, neutral, and negative signals, as well as the FinBERT classification and associated confidence score. These indicators allow the system to modulate decision thresholds when sentiment strongly supports or contradicts the forecast, while maintaining a HOLD stance when sentiment signals are weak or ambiguous. This structured display enables transparent interpretation of how sentiment contributes to final trade recommendations.

Appendix D

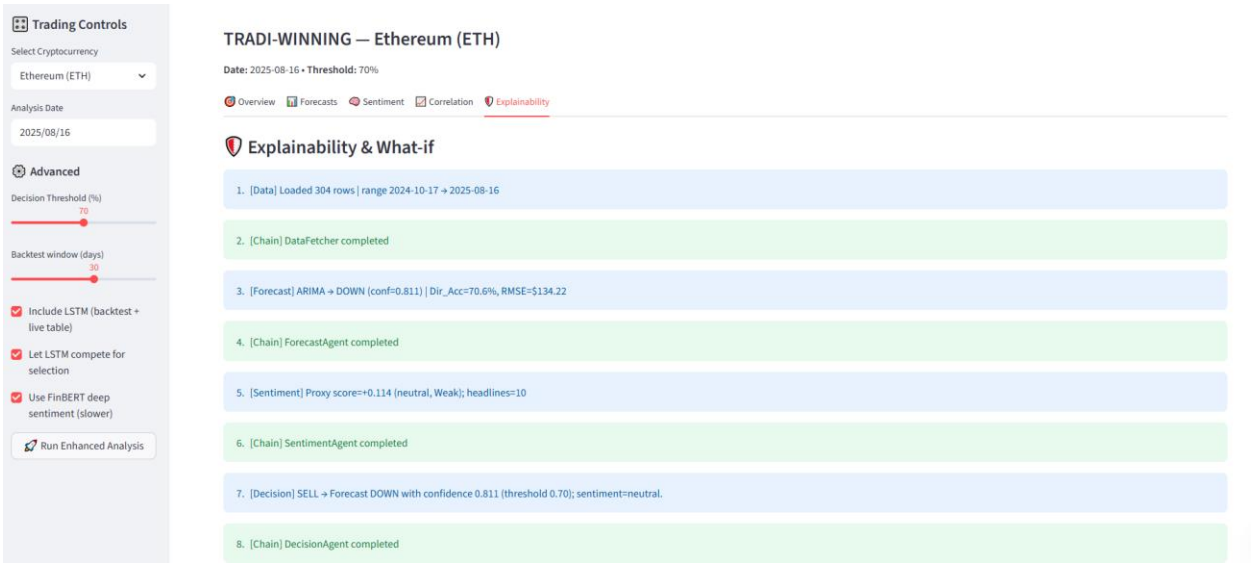


Figure 10. Explainability tab of the platform

This figure displays the step-by-step execution trace that documents how the final trading decision was produced. Each stage in the analysis pipeline: data loading, forecasting, sentiment evaluation, and decision synthesis. All of this is recorded as a discrete entry, showing the progression of information and intermediate outputs. For the example shown, ARIMA produced a downward forecast with a confidence of 0.811, sentiment was assessed as neutral, and the resulting action was a SELL recommendation based on exceeding the decision threshold. By preserving this procedural record, the platform provides a transparent and auditable reasoning chain, ensuring that each decision can be reviewed, verified, and reproduced without relying on opaque or unconstrained model behavior.

Appendix E

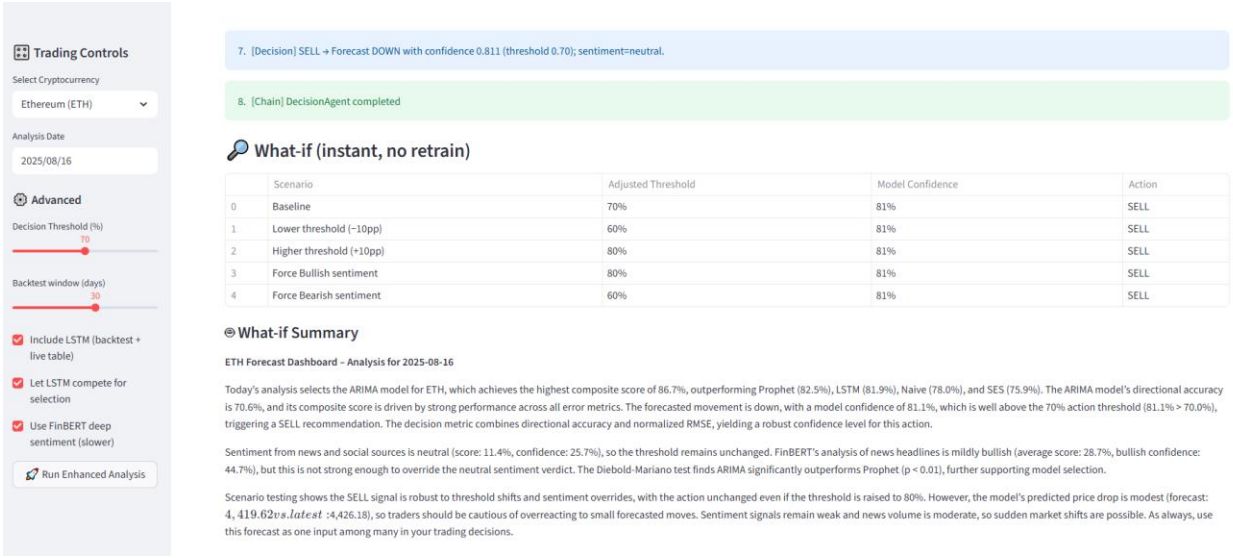


Figure 11. What-if view under the explainability tab of the platform

This figure illustrates the platform's scenario testing capability, which allows users to examine how the final trading recommendation would change under alternative threshold settings or simulated sentiment conditions. The table shows a set of instantaneous "what-if" evaluations without retraining the forecasting model, varying the decision threshold up or down and forcing either bullish or bearish sentiment. In the example shown, the action remains a SELL across all scenarios, indicating that the underlying evidence supporting the recommendation is robust to threshold and sentiment perturbations. The accompanying summary provides a narrative interpretation of these outcomes, reinforcing the transparency and auditability of the decision logic by demonstrating how sensitive (or insensitive) the final action is to key decision parameters.